GRAM: Generative Reasoning and Augmented Modeling for Quantitative Macroeconomics

Feng, Zhigang[*] Miao, Jianjun[†] Xie, Haoxuan[‡] Zhong, Yaolong[§] Zhou, Yingli[¶]

June 17, 2026

Abstract

Large language models reason fluently about economics, but their generative reasoning is bounded by what they memorised at training time and by how a query routes their attention. GRAM augments that reasoning with structured retrieval over the field's own literature. Each paper in a curated corpus is read against *PromptEcon* — a fixed basis for describing any structural macro model, organised as 3 blocks, 9 categories, and 51 named axes linked by typed edges — and the resulting per-paper instances, together with a semantic citation graph, form a single retrieval substrate. Retrieving from that substrate augments the model's context before it generates: it re-ranks the model's parametric memory toward the relevant region of the field and supplies post-cutoff work it never trained on, yielding grounded, cited answers and model code. Because the basis is typed, retrieval is *condition-aware* — it surfaces a method together with the conditions under which it is valid. We run the method as a pilot study on the overlapping-generations literature and evaluate it with a blind expert-panel exam and an algorithm- and code-generation test. The approach is not specific to OLG: it is a reproducible recipe for turning any structured economics literature into a retrieval substrate for AI-augmented research.

Keywords: knowledge graph, retrieval-augmented generation, overlapping generations, computational economics, large language models.
JEL codes: C63, C68, E13, D58.

[*]University of Nebraska at Omaha. Email: zfeng@unomaha.edu.
[†]Zhejiang University.
[‡]College of Computing and Data Science, Nanyang Technological University.
[§]University of Tokyo.
[¶]Chinese University of Hong Kong, Shenzhen.

1 Introduction

Generative AI is transforming research production across fields. Sophisticated coding, algorithm design, knowledge retrieval, and expert-level question answering — tasks that once required years of specialised training — are increasingly within reach of general-purpose large language models (LLMs). Engineers at leading AI labs now spend less than ten per cent of their time writing code themselves: at Anthropic, between seventy and ninety per cent of code is LLM-generated company-wide, and senior engineers report that they no longer write code at all and instead edit model output (Nolan, 2026). Similar shifts are documented across software-intensive industries (GitLab, 2024; Cisco Systems, 2024). The implication for knowledge work is not speculative: the bottleneck of technical implementation is moving.

Quantitative macroeconomics is an especially instructive case. The historical bottlenecks of this field — designing numerical algorithms and implementing solvers — are precisely what the latest wave of AI addresses most impressively (for recent surveys, see Dell, 2025; Korinek, 2023; Fernández-Villaverde et al., 2024; Fernández-Villaverde, 2025). If the field does not embrace this revolution, it faces a growing productivity gap relative to adjacent technical fields that have already integrated AI assistance into their workflows. This paper uses quantitative macroeconomics as an illustration of how AI is changing how we do research, and asks what infrastructure is required to make that change productive rather than merely disruptive.

The bottleneck, however, is not merely coding. General-purpose LLMs are powerful for algorithm design and implementation in domains where knowledge is broadly distributed. In medicine and law, the comparable infrastructure challenge is confidential data: practitioners assemble proprietary records, make them machine-readable, and augment a general-purpose LLM with retrieval-augmented generation (RAG) (e.g., Xiong et al., 2024). In quantitative macroeconomics the challenge is different and less recognised: the knowledge is not confidential, but it lacks the machine-readable *structure* that would make LLM augmentation effective. The field's prerequisite chains, computational bottlenecks, and solution methods are precise and formally specified — but they exist only as unstructured text in PDFs, transmitted through informal mentor–student chains. A general-purpose LLM cannot navigate a dependency graph it has never been shown; it cannot distinguish the computational concept that motivates a method from the method itself; it hallucinates method recommendations for novel problem configurations. *The information exists; the machine-readable representation does not.*

We choose overlapping-generations (OLG) models as the concrete case study for three reasons. First, the computation of large-scale OLG models — particularly with aggregate uncertainty — remains an unresolved technical challenge at the research frontier, making this an active and demanding domain. Second, the theoretical structure of OLG models is unusually rich and explicitly specified, spanning equilibrium theory, dynamic inefficiency, incomplete markets, and global numerical methods — as Weil (2008) documents in a narrative survey on the occasion of the model's fiftieth anniversary — which makes the schema design both tractable and demanding.

Third, we bring extensive research experience in this specific field, which enables credible validation of the extracted knowledge structures and the downstream research assistance the infrastructure provides; Section A surveys the curated core OLG literature in depth.

What this paper does. We build a structured knowledge graph for quantitative macroeconomics. The infrastructure has three layers — a per-paper micro anatomy, a cross-subfield macro typology, and a domain-typed knowledge graph with applicability-conditioned retrieval — instantiated on an assembled dataset of OLG papers whose core members are surveyed in detail in Section A.

Micro anatomy. Each paper is extracted as a hierarchical *concept tree* whose parent–child edges encode *derivation*, not categorical grouping. Each node carries a canonical `concept` name (the cross-paper merge key), a paper-specific prose `description` (the only field embedded for retrieval), and an optional `specifics` field that holds equations, parameter values, and notation use — segregated from the prose so that the embedding remains semantic. Two small typed tags refine each node: a binary `status` distinguishes *premises* (taken as given) from *findings* (derived); an optional `component` $\in$ {agent, firm, equilibrium, government, computation} signals where in the model architecture the concept lives. Components are tags, not scaffold nodes: they refine retrieval rather than impose an organisational hierarchy. A paper-level `notation` glossary supports the model's reading of equations across nodes. The full schema is specified in Section B.

Macro typology. Across subfields of quantitative macro, the field's compositional structure — model class $\rightarrow$ equilibrium concept $\rightarrow$ solver class $\rightarrow$ numerical refinement $\rightarrow$ diagnostic — is captured by a small *typology* of structural conditions: horizon, markets, uncertainty, agents, equilibrium concept, state-space dimension, and aggregate shocks (Section 2.5, Table 2). For the OLG depth pilot, the same vocabulary instantiates the four-era organisation surveyed in Section A — pure-exchange foundations, structural pathologies, the quantitative turn, and the deep-learning frontier — and supplies the controlled vocabulary that cross-paper edges in the graph carry. The methodology is field-general; the OLG instantiation provides the depth at which it is tested.

Knowledge graph and GraphRAG. The per-paper trees are integrated through their canonical `concept` nodes, producing a graph in which the same idea expressed across multiple papers maps to a single node carrying paper-specific descriptions side by side (Section 3.4). Each node carries the component and arc overlay of Section 3.2, so the field-level seven-slot typology that distinguishes admissible solver classes is a property of the graph itself rather than a separate retrieval-time filter. The retrieval-time protocol follows the HippoRAG pattern: a query is parsed into named entities, those entities are linked to seed nodes in the graph, Personalised PageRank propagates probability through the typed graph, and the resulting node scores are aggregated back to chunks that are passed to the LLM with a citation requirement (Section 3.4). The system does not cross the *identification boundary*: it retrieves condition-compatible methods and results, but it does not determine the equilibrium concept, state variables, or market structure of the researcher's model — that identification step remains with the researcher and is the precondition for the retrieval layer to function correctly. The full pipeline — corpus assembly, source extraction, per-paper tree construction, cross-paper merge, and the retrieval protocol — is specified end-to-end in Section 3 and instantiated on the OLG pilot.

RQ1 (schema fidelity). Does the tree-s5 schema yield a faithful, machine-readable representation of a paper's structural content, and a stable canonical-concept vocabulary across the corpus? We test this two ways. First, on a held-out subset of papers, a blind expert review compares the extracted concept tree against the structural dependencies the expert identifies in the source paper, scored on canonical-concept coverage and parent–child correctness. Second, as the corpus is processed paper by paper, we track the rate at which each new paper introduces new canonical concepts versus merging into existing nodes, and report whether the rate declines as the corpus grows — the empirical signature of a finite, learnable field vocabulary.

RQ2 (which pillar matters most?). The infrastructure rests on three pillars: paper selection (which papers to include), feature extraction (how accurately to populate the anatomy schema), and graph construction (how to assemble and traverse the resulting graph). Which of these pillars drives the largest marginal gain in end-to-end retrieval performance? Is the answer task-dependent? This question is not merely internal to our project — it provides a guideline for the emerging literature on LLM-augmented research infrastructure, indicating where effort yields the highest return.

RQ3 (domain-typed graph vs. baselines). Does a domain-typed knowledge graph outperform naive LLM prompting (Condition C1) and generic GraphRAG (Condition C3) on realistic research tasks? If so, is the gap concentrated in simple tasks, frontier tasks, or the intermediate-complexity tier?

Roadmap. Section 2 explains how an LLM mechanically produces output for the four canonical quantitative-macro research-aide tasks (model setup, algorithm design, code, discussion), why fluent local concept knowledge is not the same as operational structural reasoning, what kinds of context can repair the gap, and how four mechanisms for supplying that context — in-context stuffing, chunked RAG, generic GraphRAG, and a domain-typed GraphRAG— compare on a single running query; it closes with the cross-subfield typology of structural conditions and the argument that a domain-typed graph built from a careful reading of the literature is the right primitive for the dependency stack. Section 3 describes graph construction — extraction, embeddings, ontology induction, cross-paper edges and applicability conditions, fusion, the literature network, and the Theorem-Condition Checklist (TCC) retrieval protocol. Section 4 reports data, infrastructure, and the researcher-facing interface. Section 5 reports the evaluations. Section 6 concludes. Section A surveys the curated core OLG literature; Section B states the full schema specification; Section C traces an end-to-end pipeline run.

2 How LLMs work for quantitative-macro research — and what makes them work

A quantitative-macro researcher who turns to a large language model (LLM) for help is typically asking it to do one of a small set of things: write down a model from a verbal description, define the equilibrium concept, design a numerical solution algorithm, write runnable code, or interpret quantitative findings. This section explains what the LLM is mechanically doing when it answers, why fluent output is not trustworthy output, and what kinds of context repair the gap; we then compare three mechanisms for supplying that context — in-context stuffing, chunked retrieval-augmented generation (RAG), and generic GraphRAG — on a single running query (Section 2.4), and find that even the strongest of them surfaces the right neighbourhood of methods but cannot filter inadmissible ones. The paper's contribution, a domain-typed graph that imposes the missing applicability filter, is introduced in Section 2.5.

The thesis is one sentence: knowing the language an economist speaks is not the same as understanding how they think.

2.1 What an llm does, and what it "sees" of the literature

A frontier LLM is, at base, a conditional next-token predictor (Vaswani et al., 2017; Brown et al., 2020):

$$p_\theta(t_{n+1} \mid t_1, \ldots, t_n). \tag{1}$$

Here each t_i is a token drawn from a fixed vocabulary $\mathcal{V}$ of subword units, θ collects the trained weights of the network, and $p_\theta(\cdot \mid t_1, \ldots, t_n)$ is a probability distribution over $\mathcal{V}$ — the model's belief about which token in the vocabulary comes next. The user supplies a token sequence (the prompt); the model emits a single token, appends it to the context, emits the next, and so on. There is no symbolic engine behind the surface, no theorem prover, no compiler — only a learned probability distribution over what text "should" come next. Instruction tuning and reinforcement learning from human feedback shape the distribution, and in-context learning (Brown et al., 2020) lets the model absorb novel patterns supplied in the prompt without any parameter update; both are modifications to Eq. (1) rather than departures from it. Korinek (2023); Dell (2025) survey the implications for economic research.

What makes a model's output look like a competent economist's answer is the training distribution. Modern LLMs are trained on web-scale collections of natural-language text and source code — on the order of 10^{12} tokens drawn from public web pages, books, scientific articles, software repositories, and forum discussions (Dell, 2025; Korinek, 2023). The slice of that distribution relevant for quantitative macro contains the field's graduate textbooks, its major journal articles, lecture notes and public courseware, and its replication-code culture.[1] The model has implicitly compressed *what economists usually write next* given a similar opener. A prompt that begins "a household with idiosyncratic income shocks chooses consumption and savings to maximise expected utility" has, in the training distribution, the most-likely continuation be a Bellman equation, a budget constraint, and an Euler condition. That is what comes out.

The four canonical research-aide tasks each route through the same mechanism, conditioned on a different cluster of training trajectories. Asked to formalise a model from a verbal description, the LLM silently fills the canonical defaults the user has not specified — AR(1) idiosyncratic income, $u(c) = c^{1-\sigma}/(1-\sigma)$, Cobb–Douglas production — and lays out preferences, technology, market structure, and equilibrium definition in the standard textbook order. Algorithm design, code synthesis, and the interpretation of quantitative findings work the same way on neighbouring clusters: a known solution class is matched and its recipe recalled, an idiomatic `NumPy/Numba` template is filled, magnitudes are narrated against canonical benchmarks. In each case the model has fluent, surface-statistical access to the texts in which competent economists have produced similar outputs, and no privileged access to whether the output is structurally correct: *the model knows the language of economics; it does not know economics.*

Operational quantitative-macro questions depend on structural relationships between papers, not just on the content of any single paper. It is therefore useful to be precise about what kind of inter-paper structure the model can recover at inference time without external retrieval — that is, with no step that injects text from an external corpus into the prompt at query time, conditioning only on the user's prompt and on the trained weights θ of Eq. (1).[2] Building on the citation-function tradition (Teufel et al., 2006; Jurgens et al., 2018; Cohan et al., 2019) at the citation-intent level and on the citation-graph tradition of co-citation (Small, 1973) and bibliographic coupling (Kessler, 1963), we distinguish five paper-level relations, ordered by decreasing reliability of LLM recovery without retrieval — the first three by kind of relation, the last two by presence or absence of a formal citation edge:

1. *Lineage* — paper A extends, refines, or corrects paper B.

2. *Methodological inheritance* — paper A uses a technique introduced in paper B.

3. *Problem-tradition* — papers A and B address the same question with different tools.

[1] Concretely: graduate macro and micro textbooks (e.g., Ljungqvist and Sargent, 2018; Heer and Maußner, 2024); tens of thousands of journal articles in the major quantitative-macro outlets; lecture notes and online courses (Sargent–Stachurski's QUANTECON, Acemoglu, Krueger); public code from QUANTECON, Dynare, and replication packages distributed alongside published papers; and software-engineering material from StackOverflow and GitHub covering `NumPy`, `SciPy`, and `Numba` idioms.

[2] The weights θ are the network's parameters fixed at the end of training; they are what an LLM "knows" once deployment begins. Section 2.4 describes the retrieval mechanisms at the technical level.

4. *Citation edge* — paper A formally cites paper B, either substantively or ceremonially.

5. *Latent similarity* — papers A and B do related work but never reference each other.

For canonical papers the model recovers the first three; on the fourth it interpolates rather than retrieves, so bibliographic data comes back form-correct but frequently content-wrong, with no internal signal distinguishing genuine recall from a genre-driven guess; on the fifth it is blind, because papers that never co-occur in the training distribution leave no joint pattern for the weights to store.

What the model recovers, and why. Surveys, textbooks, and lecture notes describe lineage, methodological inheritance, and problem-tradition in prose over the canonical papers in a tradition; the model has flattened that prose into its weights and can recite the heterogeneous-agent chain Bewley–Imrohoroglu–Huggett–Aiyagari–Krusell–Smith (Bewley, 2007; Huggett, 1993; Aiyagari, 1994; Krusell and Smith, 1998), the methodological families VFI, endogenous-grid (Iskhakov et al., 2017), time iteration (Coleman, 1990), and sequence-space Jacobian (Auclert et al., 2021), and the field-level groupings (incomplete-markets heterogeneous-agent, OLG, search-and-matching, sufficient statistics) that organise them. The same prose-statistical compression is what fails on the remaining two relations: citation edges are fabricated rather than retrieved — the model interpolates plausible bibliographic data from genre statistics, and `BibTeX` synthesis comes back with the right form and the wrong content (Walters and Wilder, 2023; Algaba et al., 2025) — and latent similarity is invisible by construction, because papers that solve related problems without citing each other do not co-occur in the training distribution, and the citation-graph machinery designed precisely to surface this layer (co-citation (Small, 1973), bibliographic coupling (Kessler, 1963)) is what an LLM reading the same corpus as text does not have. Less-famous papers, working papers, dissertations, and cross-field bridges — between, say, sovereign default in an OLG economy and portfolio choice over the life cycle — inherit the same problem at the head–tail boundary (Sun et al., 2024; Kandpal et al., 2023), and substantive intellectual debt cannot be separated from ceremonial deference without reading the surrounding text (Cohan et al., 2019; Jurgens et al., 2018; Teufel et al., 2006). What the model gives back is the modal narration of the field, with the surveys' biases: recency, fame, narrative tidiness, dropped working papers, ignored siblings; the actual graph of the literature, including its non-canonical regions and its latent connections, is not what the weights store.

The implication for operational queries follows. The cross-paper *compatibility relations* that bind specifications to admissible methods are themselves tacit. They live in textbooks like Ljungqvist and Sargent (2018) that an individual research paper presumes but does not restate, and they appear in no single chunk that a similarity search could retrieve. The LLM has fluent local knowledge of every concept in the dependency stack and no graph-level access to the stack itself.

2.2 The fundamental limit: language vs. thought

Section 2.1 left the diagnosis in structural form: fluent local concept knowledge, no graph-level access to the dependency stack. The natural defense is that modern LLMs are not crude pattern matchers — during training the model develops internal representations, high-dimensional vectors that compress surface text into more abstract patterns, and these representations should let the model recover at least part of the missing structure. They are real, and they matter. When the prompt contains "OLG with CRRA utility and AR(1) idiosyncratic shocks", the model is not looking only at adjacent tokens; it activates bundled representations that link those phrases to the equations, algorithms, and code that typically accompany them in the corpus. These bundled representations are powerful, and they are what allows the model to generate plausible model statements, plausible algorithms, and plausible code from short prompts.

The decisive observation, however, is that every one of those representations was learned from surface statistics. There is no first-principles derivation under the hood, only learned associations. Bender and Koller (2020) make the in-principle case: a model trained only on form (text statistics) cannot, by that training alone, acquire reference or grounded meaning; the form/meaning distinction places a structural ceiling on what surface-statistical training can deliver. McCoy et al. (2024) carry the case empirically into the frontier-model era: even GPT-4–class systems systematically fail on deterministic tasks whose correct answer is low-probability under the training distribution, because the autoregressive next-token objective leaves "embers" in model behaviour that scaling and post-training do not extinguish. Internal representations encode the patterns visible in prose — with the same survey-author biases catalogued in Section 2.1 — but they do not encode the dependency relations *between* concepts unless those relations themselves appear in prose, and even then they are encoded as statistical associations rather than as filterable edges. The cross-paper compatibility relations of Section 2.1 live in the model as fuzzy associations between phrases — not as rules a query can apply to block an inadmissible method before the prompt is generated.

The consequence is structural. *Domain expertise is not a free byproduct of training.* It does not transfer from the training distribution to the model in a form that the model can apply at inference. The model knows the language an economist speaks but not how the economist thinks — how the language is formed out of the dependency stack of the field, how the vocabulary will evolve as the next computational era tightens the constraint set, which assumptions silently rule which methods out. The cognitive-science literature makes this distinction explicitly: Mahowald et al. (2024) draw on neural evidence that linguistic processing and reasoning occupy distinct cognitive systems and argue that LLMs exhibit strong *formal linguistic competence* (knowing the patterns of the language) but uneven *functional linguistic competence* (using language for situation modelling, reasoning, and grounded world knowledge). Surface fluency is preserved even when the recommended method is structurally incompatible with the asked-for specification, and the failure mode is silent — a phenomenon documented broadly in NLP(McCoy et al., 2019), where models that perform well by standard metrics turn out to be exploiting superficial lexical heuristics rather than the underlying competence the surface implies.

This is the structural reason a generic LLM gives confident but operationally fragile answers on quantitative-macro research questions. It is also the structural reason that retrieving more passages from the same corpus and pasting them into the prompt — the standard RAG fix described in Section 2.4 — does not close the gap: dependency relations that never appear as discrete sentences in any single paper cannot be supplied by chunk-level retrieval, however large the chunk budget. Closing the gap requires either typing the dependency relations into the context at inference time (Sections 2.3 and 2.4) or imposing them as a separate data layer on top of the corpus (Section 2.5).

2.3 Context as the repair mechanism

Two practical observations follow from the previous subsection. First, when an LLM appears to "misunderstand" a quantitative-macro question, what is happening mechanically is routing: an ambiguous prompt is mapped to the most-frequent interpretation in the training distribution, and the model emits the modal completion from that region. There is no cognitive misunderstanding under the hood, only a distributional bias toward the canonical interpretation. Razeghi et al. (2022) document this directly for numerical reasoning: model accuracy correlates strongly with the pretraining-corpus frequency of the relevant terms, which is the empirical signature of frequency-conditioned routing rather than computation. Second, the amount of context supplied at inference matters less than the *form* that context takes; sharper context routes the model to a narrower region whose modal completion is closer to the structurally correct answer.

Misunderstanding as routing to the modal interpretation. The mechanism explains a class of common failures.

- "OLG model with capital" lands the model at Diamond (1965). A user who wanted Auerbach and Kotlikoff (1987) or a modern stochastic large-cohort formulation must say so explicitly.
- "Heterogeneous-agent macro" lands at Aiyagari (1994). A user who wanted Krusell and Smith (1998) or a sequence-space Jacobian formulation (Auclert et al., 2021) must anchor the prompt to the right cluster.
- "Solve the household problem" picks VFI as the modal completion, because VFI is the most common solution method in older textbooks. The endogenous-grid method (Carroll, 2006) is more standard now in many contexts but is not the modal training answer.

The model is not missing the point; it is picking the most-likely point, which sometimes is not the user's point. Domain expertise repairs this by translating into prompts that uniquely identify the cluster the user means: canonical paper names, functional forms, algorithm class, and what to exclude. The more constrained the prompt, the smaller the manifold over which the model is interpolating, and the closer its modal completion is to what the user actually wants.

The form of context matters more than the amount. There is a hierarchy from weakest to strongest, and the qualitative gaps between adjacent levels are larger than the verbal differences suggest. Two empirical regularities make this so: LLMs recall long-tail content unreliably (Kandpal et al., 2023; Walters and Wilder, 2023), and their attention over long contexts is non-uniform with strong position effects (Liu et al., 2024) — so what the model can do with pasted text is qualitatively different from what it can do with a name.

1. *Topic words* (e.g. "OLG with capital"). Routes the model to a generic cluster.
2. *Named tradition* (e.g. "in the Aiyagari (1994) tradition"). Narrower, but relies on the model's prior about that paper, which can be inaccurate for less-famous work.
3. *Full citation* (e.g. "see Auerbach–Kotlikoff (1987), ch. 3"). Same: still recall, not content.
4. *Pasted excerpts* (equations, definitions, a methodology paragraph). Qualitatively stronger, because the model is now *reading* rather than *recalling*.
5. *The user's own partial work* (own equations, pseudocode, notation). Strongest, because it pins down the conventions every subsequent token must respect.

For canonical models on the well-trodden interior of the field, levels 1–3 are usually enough; for non-canonical territory, levels 4–5 do much more work than 1–3, and the gap between *naming* a paper and *pasting* its actual content is the largest single qualitative jump in the hierarchy.

The amplification asymmetry. A direct consequence is that LLMs amplify domain expertise rather than substituting for it. A novice asking "set up an OLG model" gets a generic two-period Diamond model — the modal answer. An expert asking the same gets a much more tailored answer, because the expert writes a sharper prompt: horizon, demographics, market structure, shocks, calibration target, solution method. The expert's knowledge is doing the narrowing; without it, the LLM falls back to whatever its training distribution makes most likely, which is usually the textbook default. The implication for a research-grade tool is that any system intended to be useful to non-experts must *supply the priors that experts would otherwise have typed in* — a pattern consistent with the field-experiment finding that AI helps most where the user has the priors to recognise where it is competent (Dell'Acqua et al., 2023). The retrieval architecture is the mechanism by which those priors are supplied.

2.4 Mechanisms for supplying context

Three mechanisms are commonly used to supply context to an LLM at inference time, alongside the no-context baseline. They differ in who curates the context (the user, an embeddingbased retriever, a graph traversal), in how much of the corpus they can carry into a single query, and in whether the supplied context preserves the structural relationships an operational quantitative-macro answer depends on. We compare them on the same running query.

> *For an OLG life-cycle economy in which overlapping cohorts face both idiosyncratic earnings risk and aggregate productivity shocks, and in which intergenerational asset markets are incomplete, which solver class can compute the joint stationary distribution of cohort wealth and asset prices in recursive equilibrium?*

In the seven-slot vocabulary that Section 2.5 introduces, this query specifies `horizon=double-sided-infinite`, `agents=cohort-indexed`, `markets=incomplete`, `uncertainty=idio.+agg.`, `aggregate_shocks=true`, `state_space_dim=moderate-high`. The structurally correct answer is a recursive equilibrium computed by set-valued Markov-equilibrium iteration on compact-valued correspondences (Krueger and Kubler, 2004, 2006; Kubler and Schmedders, 2003; Feng and Hoelle, 2017) or, when the cohort-by-asset state space is too large for sparse-grid methods, by deep equilibrium nets that learn the equilibrium policies and prices directly (Azinovic et al., 2022; Azinovic-Yang and Zemlicka, 2025; Maliar et al., 2021). Adjacent infinite-horizon methods — Krusell and Smith (1998) bounded-rationality moments, Reiter (2009) projection, sequence-space Jacobian (Auclert et al., 2021) — are not directly admissible: they assume a representative cross-section that is ergodic in the aggregate state, while the OLG setting has a cohort-asymmetric distribution that does not converge to such an ergodic object. Naively plausible but structurally wrong answers include an Auerbach and Kotlikoff (1987) backward-induction routine on a discretised cohort state (which assumes deterministic productivity and so cannot accommodate `aggregate_shocks=true`), a calibrated Auerbach–Kotlikoff descendant in the Conesa et al. (2009) mould (same deterministic-productivity foundation, repurposed for tax-policy incidence), and a perturbation around the deterministic steady state that linearises out the very welfare channel a precautionary cohort answer requires. The mechanisms below differ in how reliably they reach the correct answer rather than one of these attractors.

Figure 1 sketches the data flow of each, including a baseline "naive LLM" that supplies no external context. Table 1 summarises the qualitative properties.

(0) Naive llm. The baseline is the LLM with no external context, generating from the training distribution alone. Asked the running query, the model pattern-matches to the most-frequent OLG-policy literature in its training data — the Auerbach and Kotlikoff (1987) deterministic computational tradition and its calibrated descendants (Conesa et al., 2009) — and recommends a backward-induction routine over a discretised cohort state with a sequenced productivity path. The reply is fluent, names canonical references correctly, and is internally consistent. It is structurally inconsistent with `aggregate_shocks=true`: the recommended routine treats the productivity path as known at solve time, so cross-cohort risk-sharing is silently linearised away. The model has no mechanism for surfacing the inconsistency.

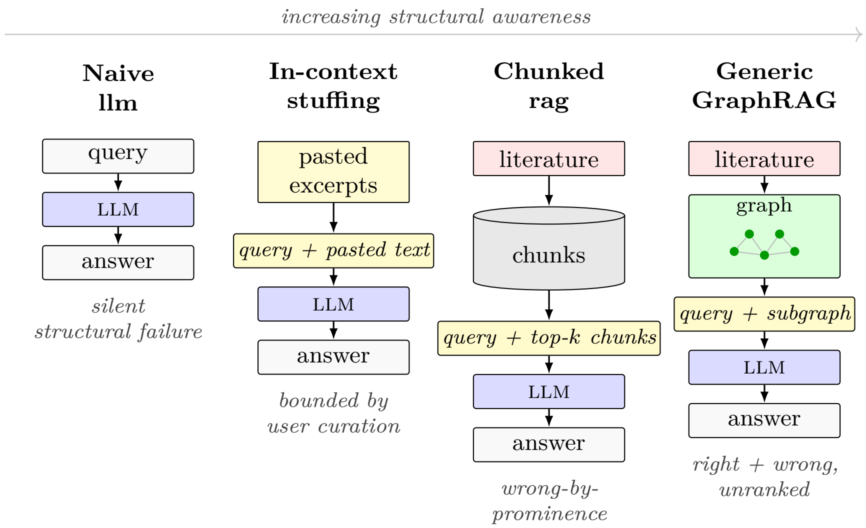

Figure 1: Four context mechanisms route the same query through progressively richer structural processing. The corpus (light pink) is the external knowledge source for chunked RAG and generic GraphRAG; in-context stuffing supplies yellow user-pasted excerpts instead; the naive LLM relies on parametric memory only. The substrate icons distinguish the storage form: a vector database of chunks for chunked RAG, and an entity–relation graph for generic GraphRAG. The yellow callout shows the augmented context the LLM sees in each architecture, and the italicised label below each answer summarises the qualitative outcome. Generic GraphRAG surfaces the right neighbourhood of methods but cannot rank correct and incorrect entries; closing that gap is the subject of Section 2.5.

(a) In-context stuffing. The user manually pastes the content the model needs into the prompt: paper excerpts, equations, prior notes, the user's own partial work. The model processes everything in a single forward pass; attention is full over the supplied context. The mechanism's strengths are full attention quality, zero infrastructure overhead, and complete user control over which material the model sees. Its limitations are also visible immediately. Coverage is bounded by the context window and by the user's curation effort: forgotten papers are invisible, and corpus-scale work (> 50 papers) does not fit. Per-query token cost is high. Long-context retrieval suffers from "lost-in-the-middle" attention degradation: a 500K-token prompt is not 500K tokens of equally-weighted attention, and the ordering of pasted material matters more than it should. Most importantly, in-context stuffing defeats discovery: it requires the user to already know which papers are relevant, which is precisely the question a research-aide tool is asked to help answer. For the running query, an expert who already knows the set-valued recursive-equilibrium literature can paste the relevant Krueger and Kubler (2004), Kubler and Schmedders (2003), and Feng and Hoelle (2017) sections, and the deep-learning extension of Azinovic et al. (2022), and obtain the correct answer; an expert is not the user this mechanism scales to.

(b) Chunked retrieval-augmented generation (rag). Lewis et al. (2020) introduced the standard pipeline: split the corpus into chunks of a few hundred tokens, embed each chunk into a dense vector with a sentence-transformer (Reimers and Gurevych, 2019; Chen et al., 2024), index the embeddings into an approximate-nearest-neighbour structure, and at query time retrieve the top-k chunks most similar to the embedded query and concatenate them into the prompt. Dense passage retrieval (Karpukhin et al., 2020) is the workhorse on which most modern RAG systems are built; a substantial variant literature studies hybrid sparse–dense retrieval, query rewriting, hierarchical summarisation (Wang et al., 2025), and domain-specialised RAG for high-stakes settings such as medicine (Xiong et al., 2024).

Chunked RAG scales to corpora of any size, and per-query token cost is small. It is a strong baseline for fact-lookup queries: the relevant chunk is retrieved, the model paraphrases faithfully, and the citation is grounded in the retrieved text. Its central limitation is that the chunk store is *flat*. Embedding similarity ranks chunks by surface proximity to the query — by which terms co-occur, not by which structural conditions are jointly satisfied. A query that requires *composing* facts across chunks — "given `markets=incomplete` together with `aggregate_shocks=true`, which solver class is admissible?" — has no chunk that states the implication, because the implication lives in the cross-paper methodology literature rather than in any single paper. The retrieved top-k may include chunks that name each individual condition, but their conjunction is unrepresentable without a structure that joins chunks by relation, not by similarity.

For the running query, chunked RAG returns top-k chunks ranked by embedding similarity. The set typically includes a Krueger and Kubler (2006) abstract, a Feng and Hoelle (2017) introduction, an Auerbach and Kotlikoff (1987) chapter passage, and a Conesa et al. (2009) calibration paragraph that pairs "aggregate productivity" with "life-cycle" at the lexical level. The chunks disagree across the deterministic / stochastic-recursive boundary, but the disagreement is structural, not verbal, and is therefore invisible to chunk-level retrieval. The LLM resolves by prominence and lands on the Auerbach–Kotlikoff–Conesa–Krueger attractor — the same wrong recommendation as the naive baseline, with strictly more text in the prompt.

(c) Generic GraphRAG. GraphRAG (Edge et al., 2024; Pan et al., 2024) replaces or augments the chunk store with a knowledge graph extracted from the same corpus. An LLM-based extractor reads each chunk and emits typed triples — entities (noun phrases) connected by relations (predicates or co-occurrence patterns) — and a separate canonicalisation pass deduplicates entities. At query time the system traverses the graph multi-hop, retrieving subgraphs whose entities and relations match the query terms; community-detection summarisation (Edge et al., 2024) aggregates results at multiple granularities, and Personalised-PageRank-based retrieval (Gutiérrez et al., 2024) ranks nodes by their importance to the query — starting from query-relevant seed nodes, importance scores propagate along graph edges, so nodes weakly connected to the query are filtered out and multi-hop neighbours of relevant seeds get surfaced. The Graphusion local-then-global pattern (Yang et al., 2024) formalises the two-step extraction

— per-document graph construction followed by cross-document fusion on canonical concepts — and hierarchical extensions support book-length corpora (Wang et al., 2025).

Generic GraphRAG dominates chunked RAG on compositional queries. The graph encodes relations no single chunk states, and traversal surfaces context that flat retrieval misses. Its limitation is that the graph is *domain-untyped.* Nodes are generic noun-phrase entities — "Bellman iteration", "aggregate shocks", "recursive equilibrium" — distinguished only by name. Edges are co-occurrence-derived relations — "mentions", "cites", "appears-with" — that record statistical association rather than structural compatibility. The extractor does not know that "Bellman iteration" is a technique and "aggregate shocks" is a condition that can silence the technique under the dependency stack of the field; nor does it know that the edge between them should be type-tagged as "method silenced under condition" rather than as generic association. The graph captures co-occurrence, not compatibility.

For the running query, generic GraphRAG reaches a broader neighbourhood than chunked RAG: "recursive equilibrium", "set-valued iteration", "deep equilibrium net", "Krusell–Smith moments", and "sequence-space Jacobian" are all surfaced via untyped multi-hop edges, alongside the "Auerbach–Kotlikoff backward induction" and "Conesa–Krueger calibration" nodes the chunked retriever already had. The graph is no longer missing context — it surfaces the structurally correct method nodes (set-valued correspondence iteration in the Krueger and Kubler (2004)–Kubler and Schmedders (2003)–Feng and Hoelle (2017) cluster, deep equilibrium nets in the Azinovic et al. (2022)–Azinovic-Yang and Zemlicka (2025)–Maliar et al. (2021) cluster) alongside the incorrect ones. But the edges carry no applicability tags. The graph cannot silence the "Auerbach–Kotlikoff backward induction" edge under `aggregate_shocks=true`, nor downweight the Krusell and Smith (1998)–Reiter (2009)–Auclert et al. (2021) edges under `agents=cohort-indexed` (which breaks the cross-section ergodicity those methods presume). The failure mode shifts from wrong-method-picked to correct-and-incorrect-listed-without-ranking — strictly better than the chunked-RAG answer, still operationally unsafe. Closing this last gap requires a typing layer that the corpus does not emit on its own; that construction is the subject of Section 2.5.

Two orthogonal axes organise the comparison: external retrieval (chunked RAG versus the naive LLM and in-context stuffing) and graph composition over the retrieval substrate (generic GraphRAG versus chunked RAG). Table 1 summarises the resulting properties. A practical decision rule emerges. For a single query whose relevant set is small and pre-curated, in-context stuffing is the right mechanism — full attention beats every retrieval substrate. For a large corpus and a content-local query, chunked RAG scales where in-context cannot. For a structural query that requires composing relations across documents, generic GraphRAG dominates: it surfaces the right neighbourhood of methods alongside the wrong ones. What none of the four delivers is condition-aware retrieval — the property that filters inadmissible methods before the LLM ever reads them. Closing that gap requires extra structure that the corpus does not emit on

Property	Naive LLM	In-context stuffing	Chunked RAG	Generic GraphRAG
Fact lookup	Partial	Partial	Yes	Yes
Compositional retrieval	No	Partial	Partial	Partial
Dependency-aware retrieval	No	User-supplied	No	Partial
Condition-aware retrieval	No	No	No	No
Scales to large corpora	—	No	Yes	Yes
Reproducible by re-running	No	Partial	Yes	Yes
Algorithm transfer	Weak	Moderate	Weak	Moderate

Table 1: Qualitative comparison of the four context mechanisms on the running query of this section. "User-supplied" means the property holds when the user has already curated the relevant papers into the prompt, which presupposes the expertise the system is meant to amplify. The condition-aware-retrieval row stays at "No" across the four mechanisms; closing that row is the contribution stated in Section 2.5, and is what quantitative results in Section 5 measure.

its own; constructing it is a research, not a data-engineering, exercise, and the next subsection states what that construction is.

2.5 Building a domain-typed graph

Section 2.4 ended with a graph that surfaces structurally relevant methods but cannot filter inadmissible ones. Closing that gap requires three artefacts the corpus does not emit on its own: a small vocabulary of applicability conditions, a per-paper anatomy that populates it, and a typing of edges between method nodes and condition nodes. This subsection sketches the architecture and what makes the typing layer non-trivial; Section 3 works through the construction in detail.

Architecture in one paragraph. Domain-typed GraphRAG extends generic GraphRAG along two axes. *Nodes* carry role and component tags from a per-paper anatomy schema (Section 3.2): each paper, however elaborate, decomposes into a small number of repeating architectural pieces — household problem, production technology, market-clearing condition, fiscal block, numerical method, and the propositions that connect them. *Nodes* also carry the field-level seven-slot specification typology that distinguishes admissible solver classes, so the applicability conditions are a property of the underlying graph. Retrieval (Section 3.4) follows the HippoRAG pattern: the query is parsed into named entities, the entities are linked to seed nodes, Personalised PageRank propagates probability through the typed graph, and the resulting node scores are aggregated back to the chunks the LLM reads. Because the typology is baked into the graph rather than imposed at query time, methods incompatible with the inferred specification have low PPR mass and are crowded out of the ranking before the LLM sees the prompt. The construction follows the Graphusion local-then-global pattern (Yang et al., 2024) with explicit role and component typing as the domain-specific extension.

Condition family	RBC / NK	Bewley–Aiyagari	HANK	OLG
`horizon`	infinite	infinite	infinite	double-sided-infinite
`markets`	complete	incomplete	incomplete	depends
`uncertainty`	aggregate	idiosyncratic	idio. + agg.	depends
`agents`	representative	continuum	continuum	cohort-indexed
`equilibrium_concept`	competitive (linearised)	stationary	ARE / SSJ	stationary, RME, ARE
`state_space_dim`	low	moderate	moderate (after closure)	moderate–high
`aggregate_shocks`	true	false	true	depends
Admissible solver class	perturbation, projection	EGM, non-stochastic simulation	sequence-space Jacobian, KS-style moments	set-valued / finite-Markov, AK-style backward induction
Structural peculiarities	—	wealth-distribution as state	sequence-space regularity	indeterminacy, dynamic inefficiency, double infinity

Table 2: The same condition vocabulary spans subfields. Each column instantiates the conditions differently and inherits different binding constraints on the admissible solver class. The OLG column is worked out in detail in Section A. *Notation*: ARE = approximate recursive equilibrium (Krueger and Kubler, 2004); SSJ = sequence-space Jacobian (Auclert et al., 2021); RME = recursive Markov equilibrium (Citanna and Siconolfi, 2010; Duffie et al., 1994); EGM = endogenous-grid method (Carroll, 2006; Iskhakov et al., 2017); KS-style = Krusell and Smith (1998)-style; AK-style = Auerbach and Kotlikoff (1987)-style.

The seven-slot typology. The dependency structure of quantitative macro is captured by a small vocabulary of applicability conditions: `horizon`, `markets`, `uncertainty`, `agents`, `equilibrium_concept`, `state_space_dim`, and `aggregate_shocks`. Each takes a small number of values; a specification is fully described by the seven-tuple, and the admissible solver class is read off by intersection. Table 2 instantiates the vocabulary across four workhorse families. Two properties matter for retrieval: the typology is *cross-subfield* — the same seven slots discriminate among RBC/NK, Bewley–Aiyagari, HANK, and OLG, so a query phrased in slot values is one the graph can answer independently of which subfield the user thinks they are in — and the compatibility relations are *conditional*, not blanket: "Bellman iteration on a discretised state space" is admissible under `aggregate_shocks=false` and `state_space_dim` small, inadmissible otherwise. The graph stores the conditional, not the unconditional, relation.

Why automated extraction does not yield this typing. The slots and their compatibility relations live tacitly in the literature. Individual papers presume the reader can navigate the dependency stack: equilibrium concept named once, solver justified in a sentence, applicability conditions left for the reader to reconstruct. The mapping from "aggregate-risk-plus-incomplete-markets-plus-continuum-of-agents" to "approximate recursive equilibrium plus Krusell–Smith-or-sequence-space methods" is taught in graduate courses and embedded in methodology references (Ljungqvist and Sargent, 2018; Judd, 1998; Heer and Maußner, 2024; Den Haan, 2010) that no individual research paper restates. An automated extractor reading the same corpus would recover the surface vocabulary — "aggregate shocks", "recursive equilibrium", "Bellman iteration" — but not the conditional rules that link them: the LLM faces the same compression problem (Section 2.2), and its output is statistical association, not filterable conditional edges. The typing has to be *designed* by a researcher reading the literature mechanism by mechanism, asking at each paper which condition has just become binding, which prior method has just been ruled out, and which new method the change forces. The worked OLG demonstration is in Section A; Section 3 describes the full construction pipeline.

Implications. Three points follow. First, the graph's filtering quality is bounded by careful, mechanism-level reading of the literature — not by how many papers an automated pipeline can process per hour. Construction is therefore a *research* task, not an engineering one. Second, the same construction generalises to other subfields — labour search, industrial-organisation entry-exit, urban-spatial general equilibrium, structural finance, public-finance life-cycle — but each subfield would specialise the seven-slot vocabulary to its own conditions and methods. We work OLG end-to-end because its dependency structure is the most demanding for retrieval: small changes in horizon, market completeness, or aggregate-shock specification flip which solver class is admissible, so a graph that filters correctly on OLG queries has to encode the most fine-grained conditional rules in the field. Third, the typed graph is the precondition for everything downstream: Section 3 constructs it, Section 4 reports the corpus infrastructure and researcher-facing interface, and Section 5 evaluates the context mechanisms quantitatively across a benchmark of operational quantitative-macro queries, using the OLG pilot as the depth instance.

3 Constructing the knowledge graph

Section 2.5 stated the construction in a high level: nodes carrying role and component tags from a per-paper anatomy schema, edges between method nodes and condition nodes carrying applicability conditions drawn from a seven-slot field-level typology, and a retrieval protocol that filters the graph *before* the LLM reads context. This section implements that statement into a construction. We describe the end-to-end pipeline that takes raw PDFs to structured text, lifts text into per-paper statement graphs, merges these into a global graph, and exposes the result as a query-conditioned retrieval surface, instantiated on the OLG pilot corpus. The architecture follows the Graphusion local-then-global pattern (Yang et al., 2024): per-document extraction first, cross-document fusion second, with a canonicalisation pass between them in the spirit of the Extract–Define–Canonicalise (EDC) protocol (Zhang and Soh, 2024). We extend Graphusion in two places: an explicit applicability-condition layer that tags method-to-condition edges with the seven slots of Table 2, and an explicit supervised-learning layer on top of the human-curated annotations produced during fusion (Section 3.4), bridging the current human-in-the-loop workflow toward a longer-run unsupervised limit.

The thesis is one sentence: *the schema is the research, and the extraction is mechanical*. The component and role vocabularies, the seven-slot condition typology, and the canonical concept names that link trees across papers are design artefacts that the research process produces and reviewers scrutinise. The pipeline below populates them; the schema is what makes population well-defined. The rest of the section follows the construction in order: Section 3.1 describes the OLG pilot corpus and its extraction; Section 3.2 the per-paper atomic-statement extraction (the layer that anchors every assumption, equation, and named result a paper emits); Section 3.3 the canonically-named concept tree built on top of those statements (the layer that names the clusters of statements the field would treat as a single contribution); and Section 3.4 the merged graph and HippoRAG-style retrieval surface.

3.1 Stage 1 — Corpus assembly and text extraction

The methodology is instantiated end-to-end on the literature of overlapping-generations models because the dependency stack — model class → equilibrium concept → admissible solver class → numerical refinement — is unusually crisp there, and each layer is forced by a structural pathology absent from the representative-agent and Bewley–Aiyagari branches. Section A surveys the field mechanism-by-mechanism in four eras of constraint accumulation; this subsection uses that survey as the seed of an extracted corpus and turns the seed into a body of papers in machine-readable form.

The corpus begins with a domain-expert reading rather than a query. A seed of foundational OLG works — anchored on the canonical foundations (Samuelson, 1958; Diamond, 1965), the general-equilibrium pathologies (Shell, 1971; Balasko and Shell, 1980; Geanakoplos and Polemarchakis, 1986; Tirole, 1985; Kehoe and Levine, 1990; Cass and Shell, 1983), the quantitative turn (Auerbach and Kotlikoff, 1987; Ríos-Rull, 1996; Conesa and Krueger, 1999; De Nardi, 2004; Krueger and Kubler, 2004), and the deep-learning frontier (Feng and Hoelle, 2017; Azinovic et al., 2022; Azinovic-Yang and Zemlicka, 2025; Han et al., 2021; Maliar et al., 2021; Zhang, 2025) — is selected by surveyor expertise rather than citation centrality: each seed paper is included because it discovers, characterises, or works against a binding structural condition that later results inherit. The seed set is documented and organised by structural condition in Section A. We then expand the corpus along the citation graph in two directions. A backward pass collects every paper the seed cites, anchoring the seed in its own intellectual lineage. A forward pass collects papers that cite the seed and are published in a fixed list of leading economics journals,[3]

[3]Forward-pass allowlist: QJE, *Econometrica*, AER, JPE, RESTUD, together with a curated set of further top-ranked economics journals — general-interest second-tier outlets and leading field journals — recorded with the

capturing where the seed has been taken up by the discipline's main outlets.

Identifiers from both passes are resolved against CROSSREF (primary) and Semantic Scholar (fallback), with a Jaccard score on title metadata serving as the accept/reject criterion.[4] A manifest tracks each accepted paper's canonical identifier, bibliographic metadata, and acquisition status, and is the single source of truth that downstream stages read. The pipeline is deliberately idempotent: each stage reads only the manifest plus its predecessor's output, so re-running a stage with revised inputs reproduces the rest of the construction without manual intervention.

The corpus contains two kinds of PDFs. The modern majority carry a text layer that can be extracted directly; a smaller set, mostly older scans, are image-only and require optical character recognition. We run all PDFs through a single extraction tool that handles both cases — it detects the document layout, parses tables and figures, applies OCR where no text layer is present, and emits a Markdown file with LaTeX-coded equations preserved.[5] The Markdown is the input that Stage 2 (Section 3.2) reads when it asks each paper to declare its model assumptions, equilibrium conditions, and unique contribution as a directed graph whose structure the equations themselves dictate.

3.2 Stage 2 — Per-paper statement extraction

Dataset framing. The OLG pilot corpus assembled in Section 3.1 is a large literature dataset built outward from a carefully selected core of foundational OLG works. We turn each paper into a directed acyclic graph of formal statements via a three-stage extraction pipeline of our own design,[6] producing per paper a graph whose nodes are statement-level units and whose edges encode logical dependency.

Per-paper DAG as the unit. Each paper contains formal mathematical content — theorems, definitions, assumptions, numbered equations, named algorithms — that must be extracted and structured before any cross-paper operation can begin. We decompose each paper into a directed

citation-expansion artefact. Restricting the forward pass to top-ranked outlets keeps the expanded corpus anchored in venues that share an editorial bar.

[4]The Jaccard score measures token overlap between the title returned by the API and the title in the request, normalised by the size of the union; a moderate threshold (we use 0.55) admits the typographic and abbreviation variations common in older economics references — title-case differences, journal abbreviations spelled out, accented characters dropped — while still rejecting matches whose titles share only generic vocabulary. Matches below the threshold are routed to manual review.

[5]The extraction runs on a remote GPU partition, because the layout-detection and OCR models are unstable on consumer hardware. We retain a plain-text PDF extractor as a fallback that always succeeds, and a separate vision-language model on cloud GPU as a cross-check whose disagreement set with the primary tool flags equation-heavy pages for review. A small number of pre-1990 image scans without an extractable text layer remain in the corpus as collected but are dropped from extraction scope.

[6]The pipeline is inspired by the agentic decomposition of Ju et al. (2026) but departs from it substantially: we replace its mathematics-only statement schema with a ten-element economics-aware vocabulary, add a component/arc overlay that connects each DAG to the seven-slot field-level typology of Table 2, and confine the LLM's role in localisation to a single regex-induction step so that the rest of the stage is deterministic. Adaptations are documented in `matlas/matlas_replica/docs/adaptations.md`.

acyclic graph (DAG) whose nodes are self-contained mathematical statements and whose directed edges encode logical dependency: a directed edge from statement B to statement A means that understanding A requires B. A DAG rather than a tree, because a single result routinely depends on multiple predecessors — a theorem may cite both an assumption and a definition. Three per-paper stages populate the graph (Locator, Structurer, Unfold; detailed mechanics in Section D); embedding, retrieval, and canonical-vocabulary consolidation are corpus-wide concerns that operate on the union of per-paper DAGs and live in Section 3.4.

Statement types. The unit of extraction is a *statement-level unit*: the smallest self-contained piece of formal content a paper introduces. We use a ten-element type vocabulary adapted for economics: `theorem`, `proposition`, `lemma`, `corollary`, `definition`, `assumption`, `claim`, `result`, `equation`, and `algorithm`. The vocabulary is deliberately broader than the pure-mathematics tradition, which typically recognises only theorem, lemma, and definition. Economics papers elevate `assumption` to a first-class unit (numbered sequences such as "Assumption A1, A2, . . . " anchor entire models); `equation` is a standalone retrieval target (a query about "the Euler condition in this model" should return an equation, not a theorem that mentions one); and `algorithm` captures contributions whose substance is procedural, such as the endogenous grid method of Carroll (2006). Each DAG node carries an identifier, a type from this vocabulary, the full statement text with mathematical notation preserved, a list of dependency identifiers, and after the Unfold stage, a self-contained restatement that can be read without consulting any other node.

The three per-paper stages, in brief. The *Locator* answers a localisation question: it asks the LLM to read the paper's equation-bearing Markdown (produced by marker-pdf in Section 3.1) and induce 3–10 document-specific regular expressions that match the paper's typographic anchors — bolded theorem headings, bracketed assumption labels, equation-number markers, algorithm blocks. The patterns are applied mechanically with `re.MULTILINE`; surviving spans are deduplicated and assigned sequential identifiers (`s0000`, `s0001`, . . .), so that the LLM's role is confined to a single induction step and the rest is deterministic. The *Structurer* reads each candidate span in context and returns a structured record — identifier, type, statement text, dependency list. Candidates are grouped into overlapping batches with a forward context window so that a theorem citing the immediately preceding lemma is still captured when a batch boundary falls between them; dependencies the LLM hallucinates are filtered against the candidate ID set. The *Unfold* stage turns the dependency lists into a directed graph and partitions it into Kahn-style topological layers; layer 0 holds every statement with in-degree zero (typically the primitive assumptions and standalone definitions); each subsequent layer is processed in order, with the LLM given each target statement together with the already-unfolded text of its *direct* parents and asked to produce a self-contained restatement. Cycles — rare but possible when a dependency is mis-attributed — are broken by promoting the lowest-in-degree node and logging

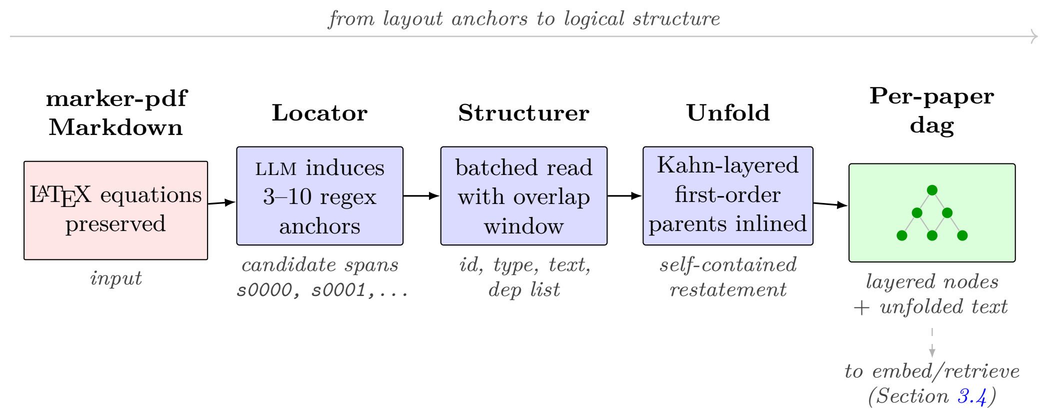

Figure 2: Per-paper Stage 2 pipeline. Equation-bearing Markdown produced by marker-pdf (Section 3.1) feeds the Locator (LLM-induced regex anchors), the Structurer (typed records with dependency edges), and the Unfold stage (Kahn-layered self-contained restatements). The output is one per-paper statement DAG; the dashed arrow marks the corpus-wide boundary — embedding and retrieval operate on the union of per-paper DAGs and are described in Section 3.4.

the break. Section D gives the full prompts, batch sizes, window widths, and dedup rules.

The Diamond exemplar. To make the pipeline concrete, consider Diamond (1965). The Locator produces 14 candidate spans; the Structurer returns 14 statements (2 assumptions, 3 definitions, 3 equations, 6 results) linked by 20 dependency edges; the Unfold stage partitions the DAG into 9 layers. Layer 0 contains the two primitive assumptions — the two-period OLG with no bequests and the CRS production technology. Layer 1 derives the savings function and the factor-price frontier. Subsequent layers build the capital accumulation law, the steady-state characterisation, the comparative statics, and at the top, the dynamic-inefficiency result and the failure of Ricardian equivalence. By contrast, Carroll (2006) — an algorithmic paper — yields 21 statements across 13 layers, with the endogenous grid method appearing as an `algorithm` node that depends on the Euler equation and the budget constraint. Across the 24 papers processed through the full pipeline so far, statement counts range from 14 to 21 per paper (median 18), dependency-edge counts from 16 to 39 (median 28), and DAG depth from 4 to 15 layers (median 9). [DATA-DEPENDENT: update counts when the full 97-paper extraction completes.]

Economic structure overlay. The three per-paper stages produce a mathematically faithful DAG; an overlay step is what turns that syntactic object into a structurally typed one the rest of the construction can filter against, and is the field-side innovation of this paper. Each extracted statement is tagged along two *orthogonal-by-design* dimensions. The first is a *component tag* drawn from a small closed vocabulary: `agent` (household optimisation, preferences, budget constraints, savings), `firm` (production technology, factor demands, profit maximisation), `equilibrium` (market-clearing, existence, uniqueness, efficiency), `government` (fiscal policy, debt, taxes, transfers, social insurance), and `computation` (numerical methods, algorithms, approximation, convergence diagnostics). The second is an *arc tag* that locates the statement in the paper's logical progression: assumption (a primitive the paper takes as given), definition (a construct the paper introduces), condition (an applicability requirement for a later result), result (a derived proposition or quantitative finding), challenge (a structural obstacle the paper diagnoses), and technique (a method or proof strategy the paper deploys). The two axes are independent because the questions they answer are independent: a paper's logic about *what* a method does, *when* it applies, and *what* it overcomes runs independently of *which* subsystem the method belongs to. The same `computation` component therefore hosts technique (the algorithm itself), condition (its applicability requirement), and challenge (the obstacle that motivates a new algorithm); the dual axis is what lets a downstream query disentangle "which numerical method" from "under what assumptions does it apply." Reading the DAG bottom-to-top through the Kahn layers traces the paper's logical arc from primitives through model setup, equilibrium definition, characterisation, and finally computation — the skeleton that every structural paper shares regardless of subfield.

The overlay closes by mapping each paper's extracted specification to the field-level seven-slot typology of Table 2: `horizon`, `markets`, `uncertainty`, `agents`, `equilibrium_concept`, `state_space_dim`, and `aggregate_shocks`. The seven-tuple is the bridge from DAG to typology: it sits on each paper's graph as a property the retrieval surface of Section 3.4 keys against. Without it, the per-paper DAGs can be searched but not pruned by specification; with it, queries phrased as a specification tuple silence methods structurally incompatible with that specification before the LLM sees any context. Component tag, arc tag, and seven-slot annotation are together the schema move that the rest of the paper's construction rests on.

3.3 Stage 3 — Concept tree: canonically-named groupings on top of statements

Two layers, complementary anchors. Stage 2 is mathematically faithful but anchors only what the paper itself emits: every assumption, every numbered equation, every named result becomes a self-contained statement node in the per-paper DAG. What it does not do is name the *clusters* of statements that the field would treat as a single concept. "OLG capital market equilibrium" in Diamond (1965) is four statements (capital-market clearing, factor-price frontier, interest-rate map, composed-dynamics stability) that any practitioner would summarise as one named object; the endogenous gridpoint method of Carroll (2006) is five statements grouped under a single algorithm. Stage 3 builds a second layer over the statement DAG — a curated *concept tree* whose nodes are canonically-named clusters of Layer-1 statements — so the corpus carries both: mechanical, statement-level, paper-internal anchors for derivational queries (Layer 1, the matlas-style DAG of Section 3.2), and curated, concept-level, cross-paper-stable anchors for what the paper *contributes* to the field (Layer 2, the concept tree of this stage). The two layers anchor different objects at different scales, and a retrieval system that wants both — "what new technique does this paper introduce, *and* which equations ground it?" — needs both, with explicit pointers between them.

Schema. Each Layer-2 concept node carries a `concept` field giving the canonical name in standard terminology (the cross-paper merge key); a `description` field with prose-only text describing what the concept means and how the paper uses it (the dense-embedder retrieval target); a `specifics` field with the math-only content — equations and parameter values from the paper, kept separate from prose so the embedder is not confused by symbols; and the same `component` tag (one of `agent`, `firm`, `equilibrium`, `government`, or `computation`) and `status` flag (`premise` or `finding`) carried at the statement layer in Section 3.2. Two further fields close the cross-layer link: `members` lists the Layer-1 statement identifiers the concept groups, and `parents` together with `edge_conditions` encode a multi-parent DAG over concepts with typed applicability conditions on each edge — the seven slots of Table 2 populate those conditions. Earlier iterations of the schema were single-parent and carried no statement-membership pointers; the two new fields earn their place by closing two limitations of the earlier design. A result that genuinely depends on multiple premises is now expressible — Diamond's dynamic inefficiency depends on *both* "no bequest motive" and "capital market equilibrium", not on either alone — and applicability conditions ("under gross substitutability", "incomplete intergenerational asset markets") are typed at the edge rather than buried in free-text descriptions, so retrieval can prune by condition before the LLM sees the prompt.

Coverage discipline. Three principles govern which Layer-1 statements earn a Layer-2 node. First, *coverage rather than partition*: a statement may belong to zero, one, or several concepts. Most statements naturally land in one concept; a Bellman equation may legitimately appear in both "household problem" and "EGM algorithm" because both are built around it. A statement with zero memberships is *scaffolding* — a standard tool the paper uses but did not introduce — and remains queryable at Layer 1 without crowding Layer 2. Second, *statement membership earns the concept*: a Layer-2 node is justified only if its name is canonical (it appears in standard terminology that other papers would recognise) *and* its `members` list is non-empty; empty-member concepts are a smell, usually indicating the proposed concept is too abstract to ground in any specific statement. Third, *concept edges follow statement edges*: a parent-child edge between concepts $A \to B$ is valid iff some statement in A's `members` is a Layer-1 ancestor of some statement in B's `members`, or the edge captures a structural derivation that statement-level dependencies cannot express (rare; flagged for human review). The discipline keeps curation tight: across the three reference papers the pipeline was calibrated against, 49 Layer-1 statements collapse to 18 Layer-2 concepts, with the rest left as scaffolding.

Extraction. The Layer-2 extractor reads two inputs per paper: the Layer-1 statement DAG produced by Stage 2 and the original equation-bearing Markdown produced by Stage 1. Conditional on those inputs, an LLM call proposes concept clusters — a canonical name, a list of member statement identifiers, parent concepts with edge conditions — under the schema and discipline above. Because the heavy mechanical work (finding every assumption, every numbered equation, every named theorem) was already done at Layer 1, the Layer-2 extractor is comparatively cheap: roughly one LLM call per paper for the grouping, against the order of sixteen calls per paper that Layer 1 itself requires. A reviewer accepts, edits, or rejects each proposed concept; the reviewed subset feeds the supervised-classifier loop documented in Section 3.4, which over time absorbs more of the mechanical labelling and shrinks the share of nodes routed to human review.

Worked examples and cross-paper merge. Figure 3 renders the construction on Diamond (1965). The Layer-2 concept "OLG Capital Market Equilibrium" groups four Layer-1 statements — capital-market clearing per worker, the factor-price frontier $w = \varphi(r)$, the interest-rate mapping ψ, and the composed-dynamics stability condition. The concept "Dynamic Inefficiency" groups two further Layer-1 statements and has *two* parents, "No Bequest Motive" (with edge condition "no operative bequest channel") and "OLG Capital Market Equilibrium" (with condition "incomplete intergenerational asset markets"). Single-parent tree schemas could only force one of these as the parent; the `parents/edge_conditions` pair now expresses both and names the structural condition under which each parent contributes. Carroll (2006)'s 21 Layer-1 statements collapse to four Layer-2 concepts (the EGM root, the endogenous gridpoint method itself, the end-of-period value function, and the liquidity constraint via prepended zero point); the remaining statements — CRRA preferences, normalised Bellman equation, impatience condition — are scaffolding because Carroll uses but did not introduce them. Once each paper has a Layer-2 concept tree, cross-paper structure emerges by name: matching `concept` strings across papers resolve to one global node in the merged graph of Section 3.4, with each paper's `description`, `specifics`, `members`, and `notation` preserved as multi-paper entries on the merged node. Layer 1 stays paper-local; the canonical-naming discipline at Layer 2 is what makes the cross-paper graph explicit rather than emergent.

3.4 Stage 4 — Graph construction and HippoRAG-style retrieval

Stage hand-off. Stage 2 emits one statement-level DAG per paper (Layer 1) and Stage 3 emits one canonically-named concept tree per paper (Layer 2), the latter aligned to the former by `members` pointers. Stage 4 takes the union of each paper's two-layer extraction and produces a single global graph keyed on canonical concept names, indexes it into a pair of vector stores (concepts and statement chunks), and exposes a query-conditioned retrieval surface an LLM can read. The pipeline has three blocks — concept-tree merging, two-store indexing, and HippoRAG-style retrieval — shown end-to-end in Fig. 4.

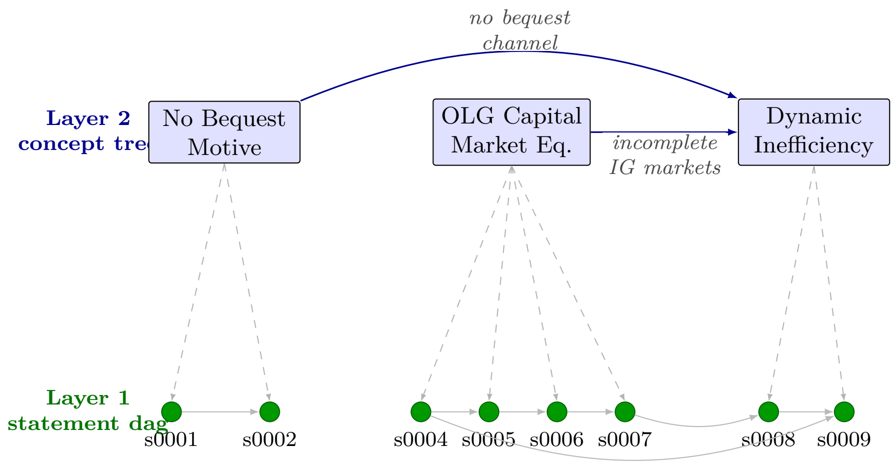

Figure 3: Two-layer architecture, illustrated on Diamond (1965). The bottom is the per-paper statement DAG from Stage 2 (Section 3.2); each green node is a self-contained mathematical statement, and grey arrows encode logical dependency. The top is the concept tree of Stage 3 (Section 3.3): each named blue node groups a cluster of Layer-1 statements via `members` pointers (dashed cross-layer arrows) and may have multiple parents with typed `edge_conditions` (solid blue arrows). "Dynamic Inefficiency" has two parents because it genuinely depends on both "No Bequest Motive" and "OLG Capital Market Equilibrium", under disjoint structural conditions that single-parent schemas could not express. Cross-paper merge happens at Layer 2 by canonical name.

Canonical labelling and concept-tree merging. Canonical concept names are produced *per paper* by the Layer-2 extractor of Stage 3, which earns them by satisfying the canonical-name discipline (P3 in Section 3.3) against the paper's Markdown and statement DAG. What remains for Stage 4 is a corpus-level validation pass plus the merge itself. The validation pass is a vocabulary builder that inventories every proposed concept name across papers, clusters near-duplicates in the embedding space, and surfaces near-collisions to a human reviewer — "recursive equilibrium" in Diamond (1965) versus "recursive competitive equilibrium" in Aiyagari (1994) is exactly the kind of pair this step is designed to catch. The reviewer accepts a single canonical spelling, rejects spurious matches, or splits genuinely distinct concepts that happen to embed close. The reviewed subset is the gold standard against which Section 5 reports extraction quality. A lightweight supervised classifier — logistic regression on frozen embeddings — trained on this reviewed subset serves error *detection* (concepts where the classifier disagrees with the extractor's proposed name are re-flagged for review) and label *propagation* (when a new paper enters the corpus, the classifier proposes canonical concept names and a human reviewer still approves additions to the canonical vocabulary). These classifiers bridge current human-in-the-loop practice toward a longer-run unsupervised limit: as the gold subset grows, the classifier absorbs more of the mechanical labelling and the fraction of concepts that require manual review shrinks.

The merge itself is a graph operation over the per-paper concept trees. Each Layer-2 node

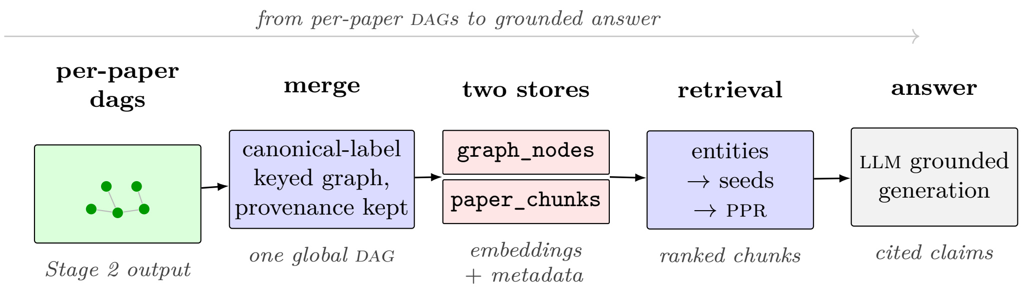

Figure 4: Stage 4 pipeline. The per-paper concept trees of Stage 3 (Section 3.3) and their grounding Layer-1 statement DAGs are merged into a single graph keyed by canonical concept name, indexed into a node store (concepts) and a chunk store (statement-level unfolded text reachable via `members`), and queried via HippoRAG-style Personalised PageRank: a query is parsed into named entities, linked to seed concepts, propagated through the graph with reverse-edge augmentation, and aggregated back to statement chunks that are passed to the generation LLM with a citation requirement.

carries a `concept` (its canonical name — the merge key), a `description`, `specifics`, `members` (Layer-1 statement identifiers), `parents` with `edge_conditions`, and the paper-level `notation` glossary. The merge traverses each per-paper concept tree recursively; for every visited node the canonical name is looked up in a global table that records, for each name, the list of source papers and the per-paper bundle of $\{\texttt{description}, \texttt{specifics}, \texttt{members}, \texttt{notation}\}$. The current paper is appended to the name's paper list and its bundle appended to the per-paper records; each parent's canonical name is appended to the node's in-neighbour list with its `edge_condition` preserved before the recursion descends. Because the canonical names were validated before this step, the same name from different papers automatically resolves to one global node; the per-paper bundles preserve provenance so no merge is opaque. The output is a node-keyed table $\{\,\text{name} \mapsto \{\text{papers}, \text{per-paper bundles}, \text{typed in-neighbours}\}\,\}$ that downstream indexing and retrieval traverse. Layer 1 stays paper-local: each paper's statement DAG is preserved alongside the global concept graph and is reachable from the merged concept node via the union of its `members` sets across papers.

Two vector stores. After merging, the two layers are indexed into two separate vector stores that share an embedding model and dimension so a query vector can be compared against either without re-encoding. The *node store* holds one vector per merged Layer-2 concept, embedded from the canonical *name* together with the per-paper *description*[7] only — paper identifiers, graph neighbours, and `members` provenance travel as metadata, not as embedding input, because the node vector should encode *what the concept means*, not *where it came from*. The *chunk store* holds one vector per Layer-1 statement chunk — the unfolded statement text produced by the

[7]Per-paper descriptions are concatenated with paper-id separators when more than one paper has merged into the same concept; the embedder sees the union of how the field has used the concept, not any single paper's spelling.

Unfold step of Section 3.2 for every statement that appears in some concept's `members` list — embedded from its full text, with paper identifier, statement identifier, and the parent concept names attached as metadata. The two stores are aligned by the `members` field: each Layer-2 node points to the Layer-1 chunks that ground it, and each Layer-1 chunk knows the Layer-2 concepts whose membership it satisfies. The separation matters at retrieval time: the node store finds structural entry points into the graph ("which concepts match this query?"), while the chunk store delivers the source text the LLM ultimately reads. Embedding-model choice is a practical trade-off: `text-embedding-3-small` is the default when cost and latency dominate; `text-embedding-3-large`, `voyage-4`, or self-hosted `bge-m3` or `Qwen3-Embedding` are drop-in alternatives when retrieval quality matters more. The common pitfalls are well known: never embed the canonical name alone (it carries no semantics outside the graph), never put paper or statement identifiers in the embedding text (they should be metadata), and never mix vectors from different models or dimensions in the same collection.

Query-time retrieval: HippoRAG-style PPR. The retrieval protocol follows the HippoRAG pattern (Gutiérrez et al., 2024), adapted to the typed-graph setting of this paper. A small LLM call first extracts the named entities and key domain concepts from the query — for the OLG corpus these are typically assumptions, model objects, mathematical properties, or named methods. Each query entity is then embedded and matched against the node store; the top-k nearest nodes for each entity become the seed set. If entity extraction fails or returns nothing, direct query-to-node retrieval serves as a fallback, so the protocol degrades gracefully on under-specified queries.

Personalised PageRank (Gutiérrez et al., 2024) starts from the seeds and propagates probability through the graph. Two adaptations matter. First, edges are traversed in both directions — forward (assumption $\to$ result) and reverse (result $\to$ assumption) — because a query about a result must be able to reach the assumptions that produce it. Second, each seed receives initial probability proportional to its retrieval score weighted by *node specificity*: the inverse of how many chunks the node appears in, so general nodes that touch many papers do not dominate over specific ones that anchor a particular result. With damping factor $\alpha = 0.85$, every node in the graph receives a PPR score reflecting its structural proximity to the query.

The score on Layer-2 nodes is converted into a score on Layer-1 chunks by a simple aggregation: each concept node distributes its PPR score to the statement chunks listed in its `members` (across all papers that contributed to the merged concept), divided by the size of that union so a concept that groups many statements does not flood the ranking. Chunks are sorted by their aggregated score and the top K are fetched from the chunk store with their full unfolded text and paper provenance. An optional hybrid term adds direct chunk-vector similarity weighted by a small λ:

$$\mathrm{score}_{\mathrm{chunk}} \;=\; \mathrm{score}_{\mathrm{PPR}} \;+\; \lambda \cdot \mathrm{score}_{\mathrm{vector}},$$

with $\lambda = 0$ recovering pure graph-based retrieval and a default $\lambda = 0.2$ providing a safety net when the query carries context the graph does not yet capture. The retrieved chunks plus their identifiers and the parent concept names that surfaced them are passed to a generation LLM with an instruction to answer only from context and to cite `paper_id` and the statement identifier for every substantive claim. Because the seven-slot overlay of Section 3.2 populates the `edge_conditions` of the Layer-2 DAG, methods that are structurally incompatible with the inferred specification slots traverse low-weight edges and have correspondingly low PPR mass; they are crowded out of the ranking before the LLM sees the prompt. The graph is the data; HippoRAG-style PPR is the protocol; the merged two-layer graph, with the typology baked into the concept edges and the statement DAGs grounding every concept, is what makes the prompt structurally safe.

4 Data, infrastructure, and interface

The methodology of Sections 2 and 3 is general; the implementation reported in this paper is instantiated on a 97-paper OLG pilot corpus, processed end-to-end on UTokyo Azure (Nakamura et al., 2025), and exposed to a researcher through a project-level slash command inside Claude Code. This section describes the input data (Section 4.1), the infrastructure that turns it into a queryable retrieval surface (Section 4.2), and the interface that a researcher actually uses (Section 4.3). The decomposition is the three-step one shared by every retrieval-augmented system — input data, retrieval pipeline, generation interface — specialised here to the schema-typed graph of Section 3.

4.1 The OLG pilot corpus

The corpus begins with a 97-paper seed assembled by the authors as a domain-expert reading rather than as a citation-centrality query; each seed paper is included because it discovers, characterises, or works against a structural condition that the rest of the literature inherits, with the selection criterion unpacked in Section 3.1. The seed is then expanded along the citation graph in two directions. A *backward* pass collects every reference the seed cites, anchoring the corpus in its intellectual lineage — Samuelson, Diamond, Cass–Shell, Kehoe–Levine, and the quantitative-turn papers of the late 1980s and 1990s. A *forward* pass collects papers that cite the seed and that appear in a fixed allowlist of leading economics journals: Tier A (QJE, *Econometrica*, AER, JPE, RESTUD) plus a curated set of second-tier general-interest and leading field journals.[8] The two passes locate the seed in its ancestry and in its downstream uptake without requiring a second domain-expert reading.

[8]The forward-pass allowlist and the citation artefacts (journal-tier classifications, raw API returns, code-availability links) are documented in a companion folder of the project repository. Restricting the forward pass to top-ranked outlets keeps the expanded corpus anchored in venues that share an editorial bar.

Identifier resolution against CROSSREF (primary) and Semantic Scholar (fallback), with a Jaccard threshold on title metadata that admits typographic variation but rejects spurious matches, is described in Section 3.1. The accepted set is recorded in a single manifest that is the source of truth every downstream stage reads and updates, and that also tracks the per-stage extraction progress of each paper; counts referenced anywhere in this paper resolve through the manifest rather than through hardcoded numbers.

4.2 Building the retrieval pipeline

Three infrastructure stages turn the manifest into the merged graph and aligned vector stores of Section 3.4: the PDF-to-Markdown extraction, the per-paper DAG construction of Sections 3.2 and 3.3, and the embedding stage. All three run on UTokyo Azure (Nakamura et al., 2025), the University of Tokyo's institutional cloud-computing tenancy; earlier stages of the pipeline were prototyped and stress-tested on NCSA Delta. We describe where each stage lives, not what it does internally — the schema and algorithms are fixed in Section 3.

From pdfs to Markdown on UTokyo Azure. The corpus is delivered as PDFs. We convert each one to structured Markdown using `marker-pdf`,[9] which handles both modern PDFs with a text layer and older scans that need OCR. The conversion is run on a GPU node of UTokyo Azure (with NCSA Delta used for development testing). Output is one Markdown file per paper, with LaTeX-coded equations preserved and embedded figures extracted alongside.

Per-paper dag extraction. The Locator → Structurer → Unfold pipeline of Section 3.2 and the concept-tree extractor of Section 3.3 are stateless: each issues calls to whichever generation model the project has configured (Anthropic Claude, GPT-4o-mini, or a self-hosted alternative). We orchestrate these calls on the same server, so the Markdown corpus and the inference client share storage. Each sub-stage's output is written to disk so that any sub-stage can be re-run independently when a prompt or schema field changes. Schema-side details (statement types, dependency edges, component and arc tags) are not repeated here; they are fixed in Sections 3.2 and 3.3.

Embedding with `Qwen3-Embedding-8B`. The merged graph and statement DAGs of Section 3.4 are indexed into the two-store layout of Section 3.4: a *node store* that holds one vector per merged Layer-2 concept, embedded from the canonical name and the union of per-paper descriptions, and a *chunk store* that holds one vector per Layer-1 statement, embedded from its full unfolded text and aligned to the concept layer through the `members` field. The production embedder is `Qwen3-Embedding-8B`, self-hosted on UTokyo Azure and served behind a local API endpoint. Self-hosting matters because the same model embeds the corpus at index time and every researcher

[9] https://github.com/datalab-to/marker. The tool combines a Surya OCR stack with a layout-detection model and emits Markdown that preserves paragraphs, tables, and LaTeX-coded equations.

query at retrieval time; placing both behind a single endpoint eliminates training–serving skew and keeps the embedding service in the same compute environment as the extraction jobs. Embedding-model alternatives — `bge-m3` (Chen et al., 2024), `text-embedding-3-small`, and `all-MiniLM-L6-v2` (Reimers and Gurevych, 2019) — are run as ablation comparators in Section 5.6; this section reports only the production default.

Reproducibility: Submission scripts, environment setup, and orchestration drivers are documented in the project repository.

4.3 Researcher-facing interface

The retrieval pipeline of Section 4.2 produces a graph plus two aligned vector stores; the interface turns that artefact into something a researcher can ask a question of. We assume the graph and indices are already built — bringing them up is a batch operation that runs once per corpus refresh — and describe only the query-time path.

Wrapping the retrieval surface as an llm tool. The HippoRAG-style retrieval of Section 3.4 is exposed as a single Python entry point that takes a natural-language query and returns the structured triple a downstream LLM needs: the named entities and seed concepts the query resolved to, the top-K statement chunks ranked by their PPR-aggregated score, and the inferred specification slots from the seven-tuple of Table 2, which let the wrapper drop methods structurally incompatible with the query before any text is generated. Two model classes sit behind the entry point: an embedding model (`Qwen3-Embedding-8B`, as in Section 4.2) used identically for the corpus and for the query, and an answer LLM — Anthropic Claude, GPT-4o-mini, or a self-hosted alternative — whose choice is configurable. The system prompt sent to the answer LLM is *cite-or-decline*: every substantive claim must carry a `paper_id` and a statement identifier drawn from the retrieved chunks, and the model is instructed to refuse rather than hallucinate when retrieval has not surfaced enough evidence. The same wrapper drives the evaluation in Section 5, so researcher-facing answers and benchmark scores share a single code path.

Researcher workflow inside Claude Code. A researcher launches Claude Code in the project repository, where a project-level slash command we provide, `/graph-query`, fronts the wrapper above. Typing `/graph-query` followed by a natural-language question runs the pipeline end-to-end: entity extraction, PPR on the typed graph, chunk aggregation, and a cite-or-decline call to the configured answer LLM. The cited answer is rendered inline in the chat, and clicking through any `paper_id` citation opens the corresponding source Markdown so the researcher can verify the underlying text. Follow-up questions re-enter retrieval rather than relying on the chat's running memory — the typed-graph filter is preserved across turns because each turn re-runs the wrapper rather than extending a chat-only context.

To make the workflow concrete, consider the running query *"OLG models with aggregate risk and incomplete markets — what equilibrium concept is appropriate?"* Entity extraction returns the set {aggregate-risk, incomplete-markets, equilibrium-concept}; seed nomination matches these to nodes in the merged concept graph; PPR surfaces statement chunks from Krusell–Smith-style and Krueger–Kubler-style papers (Krueger and Kubler, 2004) as the highest-scoring evidence; the seven-tuple inferred from the query prunes representative-agent and complete-markets methods before the answer LLM is called. What the researcher reads is a recommendation (a recursive equilibrium concept under aggregate risk), an explicit checklist of the structural conditions the recommendation depends on, and the paper-and-statement citations that ground each clause. The interface is deliberately not a chatbot wrapping free-form text retrieval: it is a structured retrieval surface that lets the LLM generate text only over a typology-filtered context, which is the property Section 5 evaluates.

5 Demonstration and evaluation

We evaluate the graph on two tracks, anchored on the OLG pilot subfield. The first (Section 5.3) is a blind expert-panel exam that probes OLG structural reasoning directly. The second (Section 5.4) is an algorithm- and code-generation test that feeds into a five-agent modelling pipeline. Section 5.5 is a scoped pipeline demonstration that shows how the graph fits into an end-to-end modelling workflow. Section 5.6 ablates graph components. Section 5.7 reports failure modes observed.

5.1 Evaluation design

We compare four conditions:

C1 — Naive llm. No retrieval. Upper bound on the base model.

C2 — Chunked rag. Top-k chunk retrieval with bge-m3 embeddings over the same corpus.

C3 — Generic GraphRAG. Domain-untyped entity graph built with Edge et al. (2024)-style extraction over the same corpus.

C4 — Domain-typed GraphRAG (ours). Micro anatomy + macro taxonomy + applicability conditions + TCC.

All four conditions use the same underlying frontier LLM so that the manipulated variable is retrieval quality, not model quality.

5.2 Diagnostic visualisation of the embedding cloud

Before the four conditions are scored on E1 and E2, we run one corpus-wide diagnostic on the extraction itself. We project the full statement-embedding cloud across all per-paper DAGs

(Section 3.2) into two dimensions with UMAP (McInnes et al., 2018) and colour the points by (i) statement type, (ii) component tag, (iii) paper or cohort, and (iv) OLG macro-taxonomy branch. The resulting plots play three roles. As a *type sanity check*: if the ten statement types separate cleanly, the extraction is producing distinguishable kinds; overlap flags mis-labelled nodes for review. As a *cross-paper alignment check*: statements that cluster tightly across papers are candidates for the canonical-vocabulary consolidation step of Section 3.4. And as *reader communication*: a UMAP of OLG statements coloured by component tag or taxonomy branch is a single-figure summary of the extracted structure (Fig. 5).

5.3 Experiment E1: expert panel on structural reasoning

We present each condition with a fixed set of three questions (OLG-pilot questions; the methodology is general), each probing one of the structural peculiarities of Section A.6:

- **Q-Indeterminacy.** Given a specific OLG specification that admits a continuum of equilibria, propose a solution strategy that respects multiplicity.
- **Q-Inefficiency.** Given a calibrated OLG economy, assess whether the steady state is dynamically inefficient and, if so, what this implies for policy evaluation.
- **Q-Infinity.** Given an OLG specification with infinite horizon and overlapping cohorts, choose a truncation scheme and justify it.

Answers are blinded to source and rated by a panel of OLG-literate reviewers on a structured rubric: (i) correctness of the equilibrium concept identified; (ii) soundness of the proposed method; (iii) explicit treatment of the structural peculiarity. A panel of three to five published OLG authors plus three to five advanced graduate students is the current design; exact composition is listed in the reproducibility notes.

[DATA-DEPENDENT: panel results; one figure reporting per-question win rates with bootstrapped confidence intervals.]

5.4 Experiment E2: algorithm and code generation

The second experiment holds out one OLG specification that is *not* in the training of the underlying LLM (a recent paper or a hand-constructed spec) and asks each condition to produce:

1. a short written description of the appropriate solution concept,
2. an algorithm in pseudo-code,
3. runnable Python code,
4. a short validation plan.

Outputs are scored on a *Structural Relevance Matrix*: for each structural dimension (equilibrium concept, horizon, markets, risk structure, numerical method class) we record match vs. mismatch with the ground truth. A failure-mode taxonomy (T1–T5 theory / Q1–Q5 quantitative; catalogued in Section 5.7) is used to categorise errors.

[DATA-DEPENDENT: scoring matrix, bootstrap intervals, per-failure-mode rates.]

5.5 Pipeline demonstration

We run the five-agent modelling pipeline on one held-out OLG specification, with the graph feeding each stage:

1. **Theorist** (skill: `modelsetup`) consumes the informal specification and emits formal mathematics — equilibrium definition, FOCs, Bellman equation. The graph provides candidate equilibrium concepts pre-filtered by the conditions of the specification.
2. **Algorithmist** consumes the mathematics and emits a solution algorithm plus pseudo-code, with cost estimates. The graph provides candidate method nodes and the applicability edges that rule out mismatched methods.
3. **Programmer** consumes the pseudo-code and emits runnable Python. The graph provides a library of reference implementations tagged by structural compatibility.
4. **Reviewer** audits any stage and flags cross-stage inconsistencies. The graph provides the condition checklists that define what it means for the downstream code to be consistent with the upstream mathematics.
5. **Analyst** consumes the solved code and produces the economic analysis. The graph provides comparable figures and moment tables from neighbouring papers for benchmarking.

Fig. 6 shows one trace; the full artifacts (code, pseudo-code, reviewer notes, analyst report) live in Section C.

5.6 Ablations

We ablate three graph components and measure the effect on the E1 and E2 primary metrics:

- drop applicability conditions (keep the graph, drop the filtering);
- drop typed edges (collapse all edge types to a single `related_to`);
- drop the macro taxonomy (flatten branches).

[DATA-DEPENDENT: ablation table; which component is most load-bearing.]

5.7 Failure modes observed

[DATA-DEPENDENT: qualitative summary of T1–T5 and Q1–Q5 instances observed across the four conditions; two or three worked examples for the paper, with longer catalogues in the appendix.]

6 Discussion and conclusion

6.1 What the results say

[DATA-DEPENDENT: one paragraph stating the headline finding on E1 and E2; one paragraph connecting to the ablations; one paragraph on what the pipeline demonstration tells us about reproducibility.]

6.2 Limitations

Four limitations structure the scope of the paper.

Single subfield, pilot sample. The corpus assembled for this project consists of 97 OLG papers, and the discussion in Section A foregrounds a pilot subset chosen to make the dependency structure legible at a human-readable granularity. Systematic full-scale extraction across the remaining corpus, and extension to subfields with less clearly-defined structural peculiarities, are the natural next steps. The methodology is deliberately transferable — schema-v4 role tags, the applicability-condition vocabulary, the TCC retrieval protocol — but whether the gains carry to corpora with weaker structural spines is an open empirical question.

Extraction quality. Tree extraction is the weakest link in the chain. The supervised-learning layer of Section 3.4 is designed to surface extraction errors, but it cannot fix an error that is consistent with the training distribution. A systematic audit of a 10-paper gold standard is the first step in any extension.

Labor cost of the current pipeline. Canonical-vocabulary construction is human-in-the-loop today: a researcher reviews canonical name proposals, and in practice the per-paper amortized time is not negligible. The supervised-learning bridge of Section 3.4 reduces this cost over time, but a fully-unsupervised induction would be a significant further step.

External dependencies. The pipeline depends on external APIs (Semantic Scholar, OpenAlex) for the literature network and on a frontier LLM for extraction and inference. Reproducibility therefore includes versioning of the LLM and freezing of the retrieval date from each API.

6.3 Future work

Extensions to other macro subfields. Natural candidates are heterogeneous-agent New Keynesian models (another family with strong internal structure) and climate-economy integrated assessment models (where the `equilibrium_concept` vocabulary already exists but is largely tacit).

GraphRAG-in-the-loop co-pilot. The five-agent pipeline of Section 5.5 is one use of the graph; a more ambitious use is a live co-pilot that surfaces challenge nodes during model construction — the equivalent of a type-checker for economic modelling.

Public release. We plan to release the knowledge graph, the canonical vocabulary, and the supervised-learning classifier as a public artifact, so that other labs can point their own OLG models at the same retrieval backbone.

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A A survey of the overlapping-generations literature

This appendix is the depth pilot for Section 2.5: the OLG subfield is the corner of quantitative macroeconomics in which the methodology of the main text is instantiated end-to-end. We choose OLG because the dependency stack that Section 2.5 sketches abstractly — model class implies equilibrium concept implies admissible solver class implies numerical refinements — is unusually crisp here: each layer is forced by a structural pathology (double-sided infinity, indeterminacy, dynamic inefficiency) that does not arise in the representative-agent or Bewley-Aiyagari branches. The survey is organised so that the structural conditions of Table 2 can be read off the OLG-specific table at the close (Table 3).

The overlapping-generations (OLG) framework occupies a peculiar position in macroeconomics. It is at once the most natural way to model an economy of finite lives and the source of a sequence of structural anomalies — Pareto-improving fiat money, dynamic inefficiency, equilibrium indeterminacy, rational bubbles, sunspot fluctuations — that the representative-agent Ramsey alternative cannot generate. This survey reads the literature mechanism by mechanism, rather than chronologically, in four stages: foundations, structural pathologies, the quantitative turn, and the deep-learning frontier. The organising thread is that every pathology is a logical consequence of substituting overlapping cohorts for an infinite-lived planner, and the resulting structural conditions accumulate into binding constraints on what a quantitative researcher can compute and conclude — choice of equilibrium concept, admissibility of solution method, direction of policy response. The literature is therefore best read as a sequence of tightening disciplines on applied modelling practice.

Modern quantitative macroeconomics is organized around a small number of workhorse frameworks. The representative-agent Ramsey model — whether in its real-business-cycle, New-Keynesian, or medium-scale DSGE incarnation — compresses household heterogeneity into a single infinite-lived planner-equivalent household. It trades away birth, death, and cohort turnover in exchange for analytic and computational tractability. The Bewley–Huggett–Aiyagari tradition (Bewley, 2007; Huggett, 1993; Aiyagari, 1994) restores uninsured idiosyncratic earnings risk, keeping horizons infinite but making the cross-sectional wealth distribution an equilibrium object. The Krusell and Smith (1998) extension and the heterogeneous-agent New-Keynesian (HANK) literature that followed (Kaplan et al., 2018; Auclert, 2019; Auclert et al., 2021) retain infinite horizons while pushing the incomplete-markets skeleton into settings with aggregate shocks, monetary policy, and distributional transmission. Across all three branches, the household is still structurally non-generational: new cohorts do not arrive with independent welfare weights, the initial old are not a distinct object, and demographic transition is not an equilibrium state variable.

The OLG literature is the branch of macroeconomics that makes those suppressed objects primitive. Samuelson (1958) and Diamond (1965) founded it by replacing the infinite-lived-agent fiction with overlapping cohorts born, aging, and dying in general equilibrium, and the subsequent literature is a sustained investigation of what that substitution implies: for existence and welfare, because the commodity space and the agent space are both infinite; for equilibrium uniqueness, because indeterminacy, sunspots, and rational bubbles become endemic possibilities; for dynamic efficiency, because capital overaccumulation and intergenerational transfers cannot be evaluated without an initial old; and for applied policy, because pensions, public debt, bequests, medical spending, and demographic transition are all cohort-specific questions. The modern quantitative branch is a fusion tradition: it embeds the Bewley–Huggett–Aiyagari idiosyncratic-risk machinery inside a cohort skeleton (Ríos-Rull, 1996; Conesa and Krueger, 1999; Heathcote et al., 2009), draws on the Krusell–Smith computational apparatus when aggregate shocks interact with moving wealth distributions, and imports life-cycle estimation and policy calibration from public finance.

Citations are selective. We discuss core papers within the set of literature this study covers, including but not limited to papers this core literature cited and papers in top economic journals that have cited those core papers. The four stages of the survey correspond to four eras of constraint accumulation: Section A.1 (Samuelson and Diamond); Section A.2 (the general-equilibrium structural pathologies); Section A.3 (the quantitative turn from Auerbach–Kotlikoff to today's heterogeneous-agent OLG); Section A.4 (the deep-learning frontier with aggregate risk). Section A.5 disambiguates two adjacent literatures — partial-equilibrium life-cycle and infinite-horizon HANK — that share machinery with OLG but are not the same object. Section A.6 consolidates the structural conditions into a single typology and bridges to the AI-augmented retrieval project that motivates the rest of the paper.

A.1 Foundations: the minimal OLG and its paradoxes

OLG originates in two papers separated by seven years. Samuelson (1958) introduces the consumption-loan model; Diamond (1965) embeds it in a neoclassical production economy. Together they fix three primitives — horizon structure, equilibrium concept, and welfare criterion — that every subsequent era inherits as binding structural conditions. The three parts below treat each primitive in turn.

The Samuelson consumption-loan model. Samuelson (1958) asks how intergenerational consumption can be reallocated in an economy with finite-lived agents, no capital, no production, and perishable endowments. Autarky emerges by default: young agents hold goods the old want; old agents have nothing to offer in exchange; generations meet exactly once. Geanakoplos (2008, p. 34) states the trap in one line — "a father can benefit from his son's resources, but has nothing to offer in return" — and shows that Samuelson's remedy is structurally minimal.[10] An intrinsically worthless token, if socially accepted, sustains perpetual young-to-old transfers of goods — with the token itself handed from old to young — and strictly raises every generation's welfare. The token is fiat money; its equilibrium return is set not by preferences but by demography. With population growth rate n, the "biological rate of interest" is $r = n$, a construct determined by demographic rather than preference primitives. Cass and Yaari (1966) draw the sharper implication: the interest rate in such an economy is, in their phrase, "a completely mechanistic

[10] Allais (1947)'s *Économie et intérêt* contains a French-language precursor to the consumption-loan model. Following Malinvaud, Weil observes that the presentation was "rather complex, leading to consideration of many cases and to introduction of long formulas," so that Samuelson's pedagogical reformulation is what made the idea generative (Weil, 2008, p. 116).

construct, having no reference to markets." This is the structural source of every OLG anomaly that follows.

Inefficiency without market failure. Samuelson's more substantive discovery is normative. A competitive equilibrium in the consumption-loan model can be Pareto-inefficient without any market failure — no externality, no informational asymmetry, no missing insurance. The Lerner–Samuelson (Lerner, 1959; Samuelson, 1959) and Meckling–Samuelson (Meckling, 1960; Samuelson, 1960) exchanges of 1959–1960 immediately probe this discovery and, as Spear and Young (2023, ch. 2) document, anticipate equilibrium indeterminacy by two decades. Samuelson's own reply to Meckling already entertains the possibility that laissez-faire competitive pricing can "condemn a society to a negative interest-rate configuration," and that rational forward-looking agents must *choose* among equilibria — a prefiguration of the modern sunspot literature treated in Section A.2.

Diamond's production economy and dynamic inefficiency. Diamond (1965) closes the founding work by embedding Samuelson in a neoclassical production economy. Young households now save by accumulating capital rather than by lending to the old, and the steady-state interest rate is endogenous to the marginal product of capital. When demographic pressure drives the market rate below the golden-rule rate, the economy over-accumulates capital, yielding *dynamically inefficient* equilibrium in the now-standard sense. Pareto-improving intergenerational transfers are then feasible *in perpetuity*: a transfer toward the initial old, financed by claims on future young cohorts, reduces overaccumulation while leaving later cohorts strictly better off along the transition. Such transfers cannot, however, be evaluated without explicitly naming the initial old and the unborn cohorts whose welfare is being compared. The asymmetry is unique to OLG.[11] The representative-agent Ramsey economy contains no distinct initial old, no sequence of unborn cohorts entering with independent welfare weights, and therefore no native way to evaluate the distributional incidence of a perpetual intergenerational transfer. For this reason every policy corollary of dynamic inefficiency — pay-as-you-go social security, public debt, and Samuelsonian fiat money as equivalent instruments for implementing such a transfer — is OLG-native.

The founding papers fix three primitives that the rest of this section traces through every era of the literature: horizon structure (overlapping cohorts with a non-empty initial old), equilibrium concept (competitive equilibrium with a possibly non-singleton selection), and welfare criterion (Pareto ranking that does not pin down a unique policy recommendation). None of the three is a

[11]This asymmetry is also the structural reason why defining a social welfare function is non-trivial in OLG settings: a single intertemporal utilitarian sum requires a discount factor on unborn cohorts that the framework cannot pin down without an extra-economic value judgement. Conventional remedies—Bergson–Samuelson aggregation with explicit cohort weights, an overtaking criterion, or a Rawlsian generation-min—are all defensible, none structurally selected, and the chosen rule shapes every welfare conclusion that follows.

free parameter; each is inherited from Samuelson (1958) and Diamond (1965), and any numerical method that proceeds as if they were free is answering a different question.

A.2 OLG as general-equilibrium theory: structural pathologies

The structural feature that generates Samuelson's welfare paradox — double-sided infinity in the commodity-and-agent topology — is the common cause of the broader class of pathologies that the next two decades catalogued. The topological root produces dynamic inefficiency, which is *not* a missing-markets problem; the two-period asset menu (fiat money versus land) then governs the dimension of indeterminacy and the admissibility of rational bubbles, subject to a gross-substitutability caveat; adding a third overlapping period generates severe multidimensional indeterminacy natively, without any nominal asset; coordination on non-fundamental beliefs (sunspots) is the direct consequence of indeterminacy; and the stochastic extension preserves every pathology even once recursiveness in the wealth distribution is established, with a new *incomplete-markets indeterminacy* emerging in the stochastic case. Each link is a binding constraint on the admissible solution methods in every downstream era. The handbook-level synthesis of the indeterminacy and sunspots literature is Benhabib and Farmer (1999).

Double infinity and the reconstruction of general-equilibrium theory. The first sharp diagnosis is due to Cass and Yaari (1966), who observe that the mechanical interest-rate mechanism is a property of the commodity-agent topology, not of impatience. The definitive geometric reading comes from Shell (1971), "Notes on the Economics of Infinity," which makes the structural claim precise. An Arrow–Debreu economy has a finite-dimensional commodity space, markets close on a compact set, Brouwer's theorem applies, and the First Welfare Theorem follows by a standard summation-by-parts argument. An OLG economy has infinitely many commodities (consumption in each period of time) and infinitely many consumers (generations born forever); these infinite-dimensional spaces behave differently.[12] Spear and Young (2023, ch. 2, sec. 2.5) put the intuition in the shape of Gamow's Hilbert-hotel paradox: in an infinite economy one can always "shift over a dimension," so the compactness-based argument on which the First Welfare Theorem rests fails, and Pareto-improving intergenerational rearrangements can be constructed by relabeling. OLG demanded a reconstruction of the general-equilibrium apparatus, not a footnote to it.

[12] It is worth distinguishing the topological pathologies of OLG from the infinite-dimensional spaces of the Aiyagari model. An Aiyagari economy also features infinitely many agents (a continuum) and infinitely many commodities (an infinite time horizon), yet it recovers the First Welfare Theorem and avoids OLG anomalies. The Aiyagari model mathematically tames its infinite agent space by integrating over a population of measure 1, ensuring aggregate quantities remain finite. It tames its infinite-dimensional commodity space because every agent is infinitely-lived and therefore subject to a terminal transversality condition. This transversality condition guarantees that the present value of the economy's endowment remains bounded over the infinite horizon, restoring the compactness required for general-equilibrium proofs. In OLG, the infinity of agents is strung along the infinity of time; no single agent evaluates the entire infinite commodity space, no overarching transversality condition applies, and compactness fails.

Gale (1973) organized the deterministic results around the Samuelson-case/classical-case dichotomy that Weil later uses as a pedagogical spine. Balasko and Shell (1980) and Balasko et al. (1980) normalized prices in present-value terms, bounded the price vector using resource constraints to obtain a compact subspace via Tychonoff's theorem, and then applied a fixed-point argument to establish existence of competitive equilibrium. Wilson (1981) proved existence under slightly different assumptions, and Geanakoplos and Polemarchakis (1986) sharpened the dimensional characterization of indeterminacy and, together with the Balasko–Shell program, anchored OLG as a general-equilibrium theory whose pathologies are not numerical accidents but visible properties of the infinite-dimensional structure.

Geanakoplos (2008) sharpens the diagnosis. The infinite-horizon OLG equilibrium set is isomorphic to that of a truncated finite economy in which the date-$T+1$ market is *not required to clear*—"classical equilibria"—and by Walras's Law as many as $L-1$ of these terminal markets can carry nonzero excess demand without violating consistency in any single period. Indeterminacy and Pareto suboptimality are therefore not market failures: they are the "right answer" to the OLG welfare puzzle, in contradistinction to the "wrong answer" that attributes the pathology to missing markets or missing intermediaries (Weil, 2008). The practical corollary is that any explanation of OLG welfare failure that appeals to missing markets or missing insurance has misstated the mechanism, and any solver recommendation grounded on that diagnosis will propose the wrong class of remedy.

Fiat money, land, and two-period indeterminacy. Once dynamic inefficiency is diagnosed as structural rather than a missing-markets problem, the natural remedy is to introduce assets that transfer wealth across generations. The classical single-good two-period OLG baseline (Gale, 1973; Balasko and Shell, 1981; Kehoe and Levine, 1985) reveals a sharp dichotomy in how different asset menus resolve the pathology. *Fiat money* — an intrinsically worthless, zero-dividend token — cures dynamic inefficiency by sustaining perpetual intergenerational transfers, but it leaves the equilibrium value of the token entirely tied to self-fulfilling expectations. Indeterminacy in the two-period model is consequently one-dimensional and almost always requires the existence of valued fiat money (a nominal steady state where the inflation factor is unity and money balances $m \neq 0$). The companion result is Tirole (1985): a rational asset bubble — a positively-priced asset whose value is sustained only by the expectation of future resale — can be Pareto-improving in an OLG economy, because the intergenerational transfers implemented by the bubble crowd out over-accumulated capital and raise every cohort's welfare. The result is impossible in a representative-agent infinite-horizon economy, where standard transversality rules it out.

Land — an infinitely-lived asset paying a strictly positive real dividend — provides the contrasting diagnostic. Because land has fundamental value, it anchors expectations: introducing productive land into the two-period economy generally eliminates the hyperinflationary indeterminacy associated with fiat money and guarantees dynamic efficiency ($r > n$) (Tirole, 1985; Magill and Quinzii, 2003; Geanakoplos, 2008). Whether an OLG economy admits bubbles or one-dimensional indeterminacy is therefore not a modeling taste; it is a structural property of which assets are spanned by the menu, and it is determined long before any solution algorithm runs.

The land cure is not global. Woodford (1984) shows that if income effects are strong enough to generate backward-bending offer curves — violating gross substitutability[13] — a robust continuum of equilibria survives even in a purely real economy with productive land. Whether gross substitutability holds is itself a model specification that must be recorded explicitly, because it gates the applicability of every uniqueness argument downstream.

The three-period breakthrough. The two-period baseline makes indeterminacy look like a manageable edge case, confined to nominal assets or pathological preferences. Kehoe and Levine (1990) overturn this reading. In their three-period-lived pure-exchange economies, the simple addition of one overlapping period generates two-dimensional indeterminacy — a continuum of Pareto-efficient equilibria — even without valued fiat money ($m = 0$). The economy exhibits pervasive instability: some steady states are not approachable by any equilibrium starting nearby, and the pathology is robust to small perturbations of utility or endowments. The three-period structure is chosen for parsimony, not for generality: while the indeterminacy of the interest-rate path in a three-period, one-good model is mathematically isomorphic to the indeterminacy of initial relative prices in a two-period, two-good, two-consumer model (Balasko et al., 1980), Kehoe and Levine's three-period specification requires only five parameters against eighteen for the two-period alternative to generate the same qualitative pathologies. The intuition is that a two-period, two-good, two-consumer pure-exchange economy must specify two preference functions and two endowment vectors over two goods — a parameter count that grows multiplicatively in the cross-sectional dimension — whereas the three-period, one-good model needs only a single preference function and a single endowment profile, with multidimensional indeterminacy emerging from the temporal horizon itself rather than from cross-good or cross-consumer interaction. The structural lesson is that computational techniques validated on two-period pedagogical models cannot be safely imported into multi-period quantitative settings: extrapolating from the two-period baseline — implicitly assuming indeterminacy is confined to fiat-money anomalies — misses the generic multidimensional indeterminacy that emerges natively in models with realistic demographic horizons.

Sunspots and expectations-driven fluctuations. Once indeterminacy is established as a structural feature rather than a knife-edge anomaly, the economic content of multiple equilibria becomes the next question. Because fundamentals do not pin down a unique equilibrium path,

[13] Gross substitutability requires that excess demand for any one good rises strictly when the price of any other good rises (equivalently, all goods are net substitutes in the Hicksian sense). It is the standard sufficient condition for *tâtonnement* stability and uniqueness of competitive equilibrium in finite economies; in OLG settings it is the load-bearing assumption that rules out the backward-bending offer curves under which a continuum of real equilibria survives even with productive land.

expectations take over: coordinated extrinsic beliefs can become self-fulfilling prophecies that dictate real allocations and business-cycle dynamics. Shell's 1977 Penn workshop circulated the first formal sunspot construction in a linear OLG. Azariadis (1981) extended the construction to nonlinear dynamics and gave the field its canonical title, "self-fulfilling prophecies," and Cass and Shell (1983) supplied the comprehensive nonlinear analysis in "Do Sunspots Matter?". Spear and Young (2023, ch. 6, sec. 6.4) reconstruct what Shell labels the "Philadelphia Pholk Theorem" that this literature implicitly suggested: *failures of the welfare theorems are exactly the opening through which non-trivial sunspot equilibria enter.* The theorem is a statement about structural vulnerability rather than a specific local claim. The welfare and market-participation conditions that exclude sunspots in finite-agent complete-markets economies need not hold in OLG, and subsequent macro papers on expectations-driven fluctuations (Grandmont, 1985, among others) build on OLG scaffolding even when the applied question is not about generations. The macroeconomic force of the observation is direct: a non-fundamental coin flip becomes a self-fulfilling business-cycle mechanism precisely because the underlying OLG structure leaves room for it.

Stochastic OLG and the recursive bridge. The stochastic extension inherits every pathology above and adds its own. Lucas (1972)'s "Expectations and the Neutrality of Money" opens stochastic OLG (SOLG) and grounds the rational-expectations research program. Spear and Young's account records a difficulty that is easy to lose in textbook summaries: general SOLG economies are not made recursive simply by declaring the exogenous shock to be the state. Spear (1985) shows that, generically, multi-good two-period SOLG economies do not admit rational-expectations equilibria in which prices and allocations depend only on the current shock or any fixed finite history. Spear and Srivastava (1986) show how lagged endogenous variables can act as sufficient statistics for the history, working with probability measures rather than a single policy function, and Spear et al. (1990) extend the indeterminacy logic to stationary stochastic equilibria. This line culminates, for existence purposes, in Duffie et al. (1994): the Markov state in a non-optimal stochastic economy is not a modeling convenience but an object whose sufficiency must be established (Spear and Young, 2023, ch. 7, sec. 7.6). The applied quantitative literature long assumed — without topological justification — that the cross-sectional wealth distribution could serve as the aggregate state. Citanna and Siconolfi (2010) close that gap. For a generic class of SOLG economies, they prove that a recursive Markov equilibrium does exist with state variables given exactly by the current exogenous shock and the wealth distribution of the agents. This is the topological bridge that justifies the entire modern heterogeneous-agent macro computational architecture.

The pathologies, however, do not disappear under recursiveness. Feng and Hoelle (2017) show that in a three-period SOLG model with *incomplete* financial markets, indeterminacy ceases to be a transient initial-condition artifact and becomes a permanent engine of volatility. By computing the entire set of competitive equilibria, they demonstrate that self-fulfilling beliefs about future asset prices drive consumption and asset-price volatility that is an *order of magnitude* larger than the volatility predicted by fundamental endowment shocks alone. The operational lesson is sharp: any model specified as "OLG + aggregate risk + incomplete markets" carries a novel *incomplete-markets indeterminacy* that no solver can resolve without an explicit equilibrium-selection rule, and that rule is itself part of the model specification rather than a post-hoc computational choice. "Recursive equilibrium," "stationary equilibrium," and "approximate recursive equilibrium" cannot be treated as aliases: they are distinct concepts with different existence conditions, method implications, and diagnostics.

The structural era leaves a specific discipline on quantitative work. A solver that returns a single equilibrium in a model that admits a continuum is not approximately right; it is silently *selecting* an equilibrium without reporting that a selection took place, and any policy experiment built on that single equilibrium is ambiguous by construction. Indeterminacy therefore commits the modeller to an explicit equilibrium-selection rule (minimum-state-variable, purification, E-stability) and rules out off-the-shelf value-function iteration that presumes a singleton solution. The same constraint governs bubble admissibility, because the structural conditions under which bubbles persist are a subset of the conditions under which indeterminacy persists.

A.3 The quantitative turn: machinery and applied policy

The structural possibilities catalogued in Sections A.1 and A.2 were turned into *computable* quantitative experiments in two moves of the late 1980s and 1990s. Auerbach and Kotlikoff (1987) demonstrated that a calibrated 55-period OLG with a realistic tax code was solvable, opening tax incidence, social-security reform, and transition welfare to quantitative analysis. Ríos-Rull (1996) merged the OLG cohort skeleton with the Bewley–Huggett–Aiyagari incomplete-markets tradition and produced the workhorse framework on which the modern applied literature depends. The framework is deployed below in three application domains — Social Security and fiscal policy, wealth inequality and late-life saving, and non-stationary demographic transition — in each of which the OLG machinery is the active ingredient: flipping any structural condition inherited from Sections A.1 and A.2 typically flips the sign of the applied answer. Adjacent literatures that use life-cycle machinery without an OLG equilibrium concept are treated separately in Section A.5.

The computational leap. The publication of Auerbach and Kotlikoff (1987) marks the transition from analytic to computable OLG. Auerbach and Kotlikoff construct a calibrated 55-period OLG in which each cohort solves a realistic life-cycle problem subject to a modern tax code, and they compute transition paths between steady states after a policy shock using shooting and Gauss–Seidel methods. Spear and Young (2023, ch. 7, sec. 7.5) appraise the contribution as a calibration-and-simulation innovation on the same order as the Kydland–Prescott turn. The technological breakthrough opens a class of quantitative questions — tax incidence by cohort, social-security reform, intergenerational equity, public-debt dynamics, transition welfare — that the representative-agent Ramsey model cannot represent at all. The initial old are load-bearing objects: no reparametrisation of a representative-agent model recovers them, and no amount of aggregate-level data discipline substitutes for the cohort-specific calibration that Auerbach–Kotlikoff make routine.

The canonical reference for the modern heterogeneous-agent OLG computational architecture is Nishiyama and Smetters (2013). They formalise the algorithm that became standard in fiscal-policy quantitative work: each cohort's optimisation is solved as a Newton-type system on the household's Kuhn–Tucker conditions; the cross-sectional distribution of households over wealth, average historical earnings, and the idiosyncratic productivity shock is iterated forward by direct simulation of the optimal decision rules; and aggregate factor prices and government-policy variables are updated through a Gauss–Jacobi outer loop until rational-expectations consistency is reached on both the steady state and the transition path. The architecture is a direct extension of the Auerbach–Kotlikoff outer loop, generalised to accommodate idiosyncratic risk and the full Bewley–Aiyagari incomplete-markets state space inside each cohort's dynamic programme, and it underlies the bulk of subsequent quantitative OLG work on tax reform, Social Security, and intergenerational fiscal accounting. For a contemporaneous methodological survey targeting OLG and recursive dynamic models, see Santos (1999).

The heterogeneous-agent synthesis. The next move merges the Auerbach–Kotlikoff cohort skeleton with the incomplete-markets heterogeneous-agent tradition. Ríos-Rull (1996) writes down the modern stochastic-OLG framework in which the aggregate state is the cross-sectional distribution of wealth across ages and each cohort solves a time-dependent Bellman problem, embedding the Bewley–Huggett–Aiyagari idiosyncratic-risk tradition (Bewley, 2007; Huggett, 1993; Aiyagari, 1994) inside the OLG skeleton and producing the workhorse framework on which the applications below depend. The empirical companion is Abel et al. (1989), who argue from U.S. data that the aggregate economy is *not* dynamically inefficient at business-cycle frequencies; subsequent work has reopened the question (Weil, 2008), but the framework remains policy-relevant even when dynamic inefficiency is empirically ambiguous, since uncertainty, incomplete intergenerational insurance, and pension-solvency risk are distinct channels through which OLG speaks to policy. Weil (2008, p. 118) supplies the cleanest methodological framing: Ramsey is the no-new-agents corner of OLG, and the three application domains that follow each ask a question for which that corner returns either the wrong sign or no answer at all.

Social Security reform and fiscal policy. The Samuelson–Diamond–Tirole case for Social Security as an intergenerational insurance device is revived in the quantitative era in a form only OLG-with-heterogeneity can address. Conesa and Krueger (1999) embed a pay-as-you-go pension system inside a Bewley–Aiyagari–OLG economy with idiosyncratic labor-productivity risk and decompose welfare into the insurance value of PAYG transfers (zero by construction in a representative-agent model) and their crowding-out cost on aggregate capital. Krueger and Kubler (2006) push the analysis to a nine-period stochastic OLG, finding that a Pareto improvement from introducing PAYG requires *both* high risk aversion and a high intertemporal elasticity of substitution; under standard preferences the crowding-out channel overturns the intergenerational-risk-sharing channel. Conesa et al. (2009) extend this to optimal capital taxation: the life-cycle structure itself — borrowing constraints and the age profile of labour income — delivers a near-complete reversal of the Chamley–Judd zero-capital-tax prescription, with a Ramsey-optimal capital-income tax of roughly 36% once life-cycle heterogeneity is admitted. All three results are compositional claims — the welfare sign depends on accumulated structural conditions inherited from Sections A.1 and A.2 — and all three are unreachable from the representative-agent baseline. Handbook-level surveys covering the broader fiscal-policy and life-cycle context include Elmendorf and Mankiw (1999) on government-debt economics and Attanasio (1999) on the life-cycle consumption tradition.

Wealth inequality, bequests, and the health-cost channel. Three age-heterogeneous mechanisms together account for the right tail of the U.S. wealth distribution and the saving behaviour of the elderly — objects the representative-agent benchmark cannot match. De Nardi (2004) shows that voluntary bequests combined with parent–child earnings persistence generate the observed concentration at the upper percentiles; without either ingredient the Gini and the top-1% share collapse. Cagetti and De Nardi (2006) add entrepreneurship and collateral-based borrowing frictions, with entrepreneurial risk *and* bequest motives jointly driving the very top of the distribution. De Nardi et al. (2010) resolve the puzzle of late-life saving by introducing out-of-pocket medical-expense risk that rises with age and income, reassigning the majority of upper-quintile retiree saving from a bequest motive to a precautionary motive against medical expenditure (and, by corollary, giving Medicaid a non-negligible welfare value far above the eligibility thresholds). Chen et al. (2025) extend the health-cost channel by making coverage endogenous to health trajectories, generating a composition-of-survival *health premium* on insurance expansion that raises measured inequality in realised health, lifetime earnings, and wealth. All four channels require the OLG skeleton: they are age-heterogeneous by construction, and a representative-agent model has no age structure to exploit.

Non-stationary demographics and asset prices. A third domain in which OLG is load-bearing is the co-movement of asset prices with the cohort-size trajectory. Geanakoplos (2008) shows that, even under a perfectly anticipated demographic path, asset prices respond to the relative size of savers and dissavers in a way a representative-agent economy cannot represent. Pay-as-you-go solvency, labour-supply dynamics under aging, and pandemic-driven intergenerational transfers all hinge on the demographic trajectory itself rather than on the shocks it conditions, and any solver that substitutes a stationary-demography approximation answers a different policy question.

The quantitative turn closes the loop opened by the foundational and structural eras. Once Auerbach–Kotlikoff transition paths are tractable and the Bewley–Aiyagari merger gives them an idiosyncratic-risk skeleton, the policy questions for which OLG is strictly mandatory — not merely useful — are exactly those in which the cohort distribution is moving, the welfare incidence falls on the initial old, or the right tail of the wealth distribution is the object of interest. The methodological corollary is sharp: Auerbach–Kotlikoff shooting, Krusell and Smith (1998) bounded-rationality forecasting, and non-stochastic simulation (Young, 2010) are admissible only when the structural conditions inherited from Sections A.1 and A.2 hold; a question about Social Security, wealth concentration, or demographic transition that is routed to a representative-agent solver is mismatched at the architectural level rather than at the parameterisation level.

A.4 Frontier: stochastic OLG with aggregate risk

Aggregate risk in a large-scale OLG is structurally harder than in a representative-agent model: the aggregate state is the cross-sectional distribution of wealth over ages and idiosyncratic states, in principle infinite-dimensional. Each method below pays an explicit price in dimensionality, rigour, or equilibrium concept.

Krusell and Smith (1998)'s bounded-rationality scheme has agents forecast prices using a low-dimensional summary of the wealth distribution with consistency verified ex post; though not natively OLG, it is the backbone of OLG-with-aggregate-risk applications, supported by Young (2010)'s non-stochastic simulation and the endogenous grid method (Carroll, 2006; Iskhakov et al., 2017) for non-concave value functions. Krueger and Kubler (2004)'s *approximate recursive equilibrium* applies the same parametric-reduction strategy to the equilibrium correspondence, with error bounds in Euler-equation residuals.

The set-valued tradition grounds tractability in monotone iteration. Duffie et al. (1994) establish that stationary Markov equilibria in non-optimal economies are correspondences — the largest fixed point of a monotone operator on compact-valued correspondences — and Feng et al. (2014) translate this into an iterative algorithm that contracts a compact-valued $\mathbf{V}_0 \supset \mathbf{V}^*$ to convergence, with diagnostics on set convergence rather than a scalar Bellman residual. Kubler and Schmedders (2003) document explicit non-existence results in adjacent specifications.

The deep-learning frontier abandons grids. *Deep equilibrium nets* (Azinovic et al., 2022) approximate policies and prices by feedforward networks trained on simulated ergodic paths against an unsupervised residual loss; benchmarked on a 56-cohort OLG with a 108–292-dimensional state, mean Euler errors run below 0.025%. DEEPHAM (Han et al., 2021) learns the sufficient statistics for the cross-section automatically, and Azinovic-Yang and Zemlicka (2025) take a structurally different turn by parameterising equilibrium objects as functions of a truncated history of exogenous shocks, avoiding the endogenous-state feedback that destabilises recursive deep nets. Zhang (2025) adapts the architecture to a finite-horizon OLG with both idiosyncratic and aggregate risk and disciplines it against stock-demand inelasticity (Constantinides et al., 2002; Maliar et al., 2021; Fernández-Villaverde, 2025).

Three open problems stand out. *Theoretical*: indeterminacy survives in stochastic OLG with aggregate risk and changes character; Feng and Hoelle (2017) distinguish initial-condition indeterminacy from a more severe incomplete-markets indeterminacy that becomes a permanent engine of volatility, with disciplined equilibrium-selection rules left to future work. *Methodological*: convergence guarantees for deep equilibrium nets remain weak, and neural representation of equilibrium correspondences is essentially absent. *Applied*: non-concave value functions over several state variables (health, human capital, wealth, lifespan) and the non-stationary demographic trajectory of Geanakoplos (2008) both lie beyond the reach of present solver inventories.

A.5 Adjacent literatures

Two adjacent literatures share apparatus with age-structured OLG on applied questions without being the same object. The distinction hinges on whether the aggregate cohort distribution is a load-bearing equilibrium object: the OLG branch surveyed above treats cohorts born, aging, and dying in general equilibrium — and the aggregate age distribution they generate — as a primitive of the general-equilibrium state vector. Partial-equilibrium life-cycle models and infinite-horizon perpetual-youth devices may share mortality, survival, or age-profile primitives with OLG, but they do not place an age-resolved cohort distribution at the centre of the equilibrium state. Keeping the branches distinct matters because the admissibility of a given solution method and the admissibility of a given policy counterfactual both depend on which branch the model occupies.

Life-cycle consumption, saving, and portfolio choice in partial equilibrium. A long-standing applied tradition studies household consumption and portfolio decisions at disaggregated frequency using age-indexed dynamic programming without closing the general-equilibrium loop. Hubbard et al. (1995) supply the canonical quantitative case that means-tested social-insurance programs depress saving among low-permanent-income households. Storesletten et al. (2004) estimate the fraction of idiosyncratic consumption variability over the life cycle that is not insured. Cocco (2005) shows that housing — modeled as a lumpy, transaction-cost-heavy illiquid asset whose service flow enters utility — crowds out risky financial-asset holdings throughout the life cycle. Gomes et al. (2008) put flexible labor supply at the center of the life-cycle portfolio problem: endogenous labor supply acts as a partial hedge against financial-market shocks, rotating the optimal glide-path of equity exposure in a quantitatively large way. Low et al. (2010) decompose labor-income risk into wage risk and employment risk and estimate their separate life-cycle profiles. These papers use the life-cycle apparatus the OLG tradition helped develop — age-indexed value functions, survival probabilities, discretized shocks, non-concave optimization — but solve a single household's problem at exogenous factor prices rather than closing the general-equilibrium loop over the aggregate age distribution. Their quantitative results are not OLG results; they are upper bounds or partial-equilibrium benchmarks that OLG models in Section A.3 either reproduce, refine, or overturn when the general-equilibrium feedback is reinstated.

Infinite-horizon heterogeneous agents and hank. The heterogeneous-agent New-Keynesian literature of the last decade is sometimes described as a fusion of the OLG-life-cycle tradition with Krusell–Smith. That characterization is historically imprecise. The canonical HANK models chose the infinite-horizon Bewley–Huggett–Aiyagari branch, not the age-resolved OLG branch. Kaplan et al. (2018) add a Blanchard perpetual-youth device — a constant Poisson death hazard $\lambda = 1/180$ per quarter, so that expected life is 45 years — to an otherwise infinite-lived model; agents who die are replaced by newborns who inherit zero wealth and a draw from the ergodic productivity distribution. This creates turnover, but not the age-resolved cohort distribution that drives Auerbach–Kotlikoff or Ríos-Rull-style OLG policy analysis. Auclert (2019) works with an abstract life-cycle household in the theoretical framework and a Bewley–Huggett–Aiyagari infinite-horizon economy in the quantitative implementation. Auclert et al. (2021) use pure infinite-horizon agents with no death and no generational structure; the sequence-space Jacobian algorithm they introduce takes the representative-household Euler equation and the cross-sectional asset distribution as its primitive objects, with no role for an aggregate cohort distribution. The methodological interface between HANK and OLG runs through shared numerical infrastructure — Krusell–Smith approximate aggregation, Young (2010)'s non-stochastic simulation, sequence-space linearization — rather than through a common equilibrium concept. Whether a specific technique (e.g., sequence-space Jacobians) ports to an OLG environment is therefore not automatic; its applicability depends on features of the cohort-structured state space that HANK suppresses by construction.

A.6 Structural conditions and the methodological discipline they impose

The four eras combine into a single substantive lesson. OLG is not a niche refinement of the representative-agent Ramsey economy but a separate body of macroeconomic theory whose structural peculiarities — (i) double-sided infinity in the commodity-and-agent topology, (ii) dynamic inefficiency relative to the golden rule, and (iii) equilibrium indeterminacy in both the deterministic and the stochastic case — accumulate over the lineage and propagate downstream as binding constraints on equilibrium concept and admissible solution method. A model specified as "OLG with aggregate risk and incomplete markets" is not a neutral collection of modelling choices; it inherits double-sided infinity from Shell (1971) and Balasko et al. (1980), dynamic inefficiency from Diamond (1965) and Tirole (1985), indeterminacy from Azariadis (1981), Cass and Shell (1983), and Kehoe and Levine (1985), and a specific equilibrium concept — approximate recursive equilibrium from Krueger and Kubler (2004) or set-valued Markov-equilibrium iteration in the Duffie et al. (1994) / Feng et al. (2014) sense. The same pattern surfaces in the applications

Structural condition	**Era**	**Canonical references**	**Binding constraint on admissible method**
Double-sided infinity	Foundations	Shell (1971); Balasko et al. (1980); Geanakoplos (2008)	Existence and welfare proofs cannot rely on finite-dimensional compactness.
Dynamic inefficiency	Foundations	Diamond (1965); Tirole (1985); Abel et al. (1989)	Initial old and unborn cohorts must be named for any intergenerational transfer to be welfare-evaluated.
One-dimensional indeterminacy	Pathologies	Gale (1973); Kehoe and Levine (1985); Woodford (1984)	Selection rule (e.g., MSV, purification) is part of the model specification, not a post-hoc numerical choice.
Multidimensional indeterminacy	Pathologies	Kehoe and Levine (1990)	Two-period numerical validation does not extend to realistic demographic horizons.
Sunspot equilibria	Pathologies	Azariadis (1981); Cass and Shell (1983); Grandmont (1985)	Off-the-shelf value-function iteration assumes a singleton; explicit selection is required.
Rational asset bubbles	Pathologies	Tirole (1985)	Bubble admissibility depends on the asset menu; the menu is itself a modelling choice.
Recursive bridge in stochastic OLG	Pathologies	Spear (1985); Duffie et al. (1994); Citanna and Siconolfi (2010)	Wealth-distribution Markov state is sufficient only generically, not by declaration.
Incomplete-markets indeterminacy	Frontier	Feng and Hoelle (2017)	Self-fulfilling beliefs are first-order in stochastic OLG as well as deterministic; selection rule is part of the model.

Table 3: Structural conditions accumulated by the OLG literature, the era at which each was identified, and the constraint each imposes on admissible solution methods. Each row is a structural fact whose violation either reverses the sign of an applied answer or invalidates the convergence argument of a numerical method.

of Section A.3: the Pareto-improving PAYG, positive-capital-tax, and medical-expense-saving results each sit on a stack of structural conditions inherited from earlier eras, and flipping any single condition can flip the sign of the applied answer.

Table 3 consolidates the inherited conditions into a single typology, organised by era of discovery and canonical reference, with the binding constraint each condition imposes on admissible solution methods. The three peculiarities of the preceding paragraph are accordingly not a separate taxonomy grafted onto OLG: they are the accumulated residue of the four eras, each attached to the historical moment at which it was discovered.

The same lineage that disciplines the working economist's choice of solver also disciplines what an LLM must retrieve to produce a structurally safe recommendation. Surface fluency can describe Samuelson's consumption-loan model and can repeat that OLG admits sunspots, yet the compositional step — which structural condition rules out which solver class, and therefore which algorithm is admissible for a given model specification — requires dependency-aware retrieval that a textual language model does not natively support. The same compositional difficulty propagates to applied policy questions: whether a given Social-Security counterfactual requires bequest heterogeneity, or whether a policy exercise that crosses the OLG/HANK boundary can rely on the sequence-space Jacobian at all, is itself a compositional question over the structural conditions of Table 3. Section 2.4 compares the context mechanisms on exactly this compositional task, and the remainder of the paper argues that only a domain-typed GraphRAG can respect the dependency direction the OLG literature has spent sixty-five years establishing.

B Schema specification

This appendix states the full schema used for micro anatomy trees (schema v4) and the edge vocabulary for the macro graph.

B.1 Schema v4 — role tags

Each node in a per-paper tree carries an optional role tag drawn from six values:

assumption. A primitive postulate that the paper imposes (e.g. "utility is time-separable and CRRA"). Not derived from deeper primitives within the paper.

definition. A concept whose meaning is fixed by the paper (e.g. "recursive Markov equilibrium for this economy"). Definitions do not encode assumptions.

condition. A qualification that an object satisfies under specified hypotheses (e.g. "transversality condition"). The graph's applicability-condition vocabulary is built on the condition role.

result. A theorem, proposition, or numerical finding that the paper states or proves (e.g. "the equilibrium price supports exist and are unique under assumptions A1–A3").

challenge. A known obstruction to modelling or computation (e.g. "indeterminacy", "curse of dimensionality"). challenge nodes carry `requires` edges to technique nodes that address them.

technique. A numerical, analytical, or econometric method (e.g. "value function iteration on a Chebyshev polynomial basis", "Krusell-Smith bounded-rationality forecast").

The set is deliberately small: six roles is the point at which extraction agreement among human reviewers is high enough for the annotations to be trusted as gold labels.

B.2 Applicability conditions

`horizon.` `finite` | `infinite` | `double-sided-infinite`.

`markets.` `complete` | `incomplete` | `constrained`.

`uncertainty.` `none` | `idiosyncratic` | `aggregate` | `idiosyncratic_plus_aggregate`.

`agents.` `representative` | `heterogeneous` (with sub-field for the source of heterogeneity).

`equilibrium_concept.` `stationary` | `recursive_markov` | `approximate_recursive` | `sequence_of_markets`.

`state_space_dim.` integer or a condition (`finite`, `infinite-reduced`, `infinite`).

`aggregate_shocks.` boolean.

Each edge in the graph (within-paper or cross-paper) may carry any subset of these conditions; retrieval filters on the intersection of the query-inferred conditions and the edge conditions.

B.3 Edge types

Within-paper. Parent-of, part-of, and a small set of typed relations (`requires`, `addresses`, `uses`).

Cross-paper. `same_as`, `subtype_of`, `precursor_of`, `related_to`.

Literature network. Kept: `extends_model`, `uses_method`, `addresses_problem`, `requires_theory`, `refines_or_corrects`. Dropped: `shares_calibration`, `background_context`, `ceremonial`, `shared_technique`.

[DATA-DEPENDENT: final schema counts and a diagram of the edge-type distribution.]

C Pipeline demonstration trace

This appendix lists the artifacts produced by each agent in the pipeline demonstration of Section 5.5.

[DATA-DEPENDENT: artifacts to be committed alongside the final draft: `demos/olg_eval/description.md`, `mathmodel.md`, `algorithm.md`, `code/main.py`, `reviewer_notes.md`, `analyst_report.md`.]

C.1 Theorist output

[DATA-DEPENDENT: formal statement of equilibrium, first-order conditions, Bellman equation. Cross-reference the graph nodes consulted.]

C.2 Algorithmist output

[DATA-DEPENDENT: pseudo-code, cost estimate, justification of method class. Cross-reference the applicability conditions resolved.]

C.3 Programmer output

[DATA-DEPENDENT: `main.py` and helper modules, with a short description.]

C.4 Reviewer notes

[DATA-DEPENDENT: cross-stage consistency checks, any findings routed back to earlier agents.]

C.5 Analyst report

[DATA-DEPENDENT: figures, moments, counterfactuals, comparison with benchmark papers identified via the literature network.]

D Per-paper extraction pipeline: detailed mechanics

This appendix gives the full mechanics of the three per-paper extraction stages summarised in Section 3.2. The architecture is inspired by the agentic decomposition of Ju et al. (2026); we cite our adaptations inline where they apply, and a complete adaptations register lives in `matlas/matlas_replica/docs/adaptations.md`.

D.1 Stage 1: Locator

The Locator answers a localisation question: where in the document do statements live? Different papers number theorems differently, format definitions differently, and use different naming conventions. A single LLM pass over the full document could guess where one paper's "Theorem 3.1" starts, but it cannot do so reliably across every paper without first seeing each paper's local conventions. Our solution is a two-step design: the LLM reads the paper's equation-bearing Markdown (produced by marker-pdf in Section 3.1) and *induces* a set of document-specific regular expressions that match this particular paper's typographic anchors — bolded theorem headings, bracketed assumption labels, equation-number markers, subsection headers that name results, algorithm blocks. These 3–10 regex patterns are then applied mechanically via `re.MULTILINE`: each pattern is compiled, matched against the full Markdown, and the resulting spans are sorted by document position and deduplicated against overlaps (the first match is kept when two patterns fire at the same location). The surviving candidates receive sequential identifiers `s0000`, `s0001`, ... and are passed to Stage 2. The design confines the LLM's role to a single induction step; everything after it — pattern application, deduplication, ID assignment — is deterministic, fast, and reproducible. Invalid regex expressions produced by the LLM are caught at compile time and silently skipped; the remaining patterns continue.

D.2 Stage 2: Structurer

The Locator produces candidate spans; the Structurer reads each candidate in context and extracts a structured record. Candidates are grouped into batches of five and each batch is presented to the LLM together with a forward context window of 4,000 characters starting at the first candidate's position and extending past the last candidate by half a window width. The overlap is the critical design choice: a theorem that cites the immediately preceding lemma will be captured even when a batch boundary falls between them, because the overlapping window ensures both are visible in the same LLM call. For each candidate the LLM returns a structured record — identifier, statement type, full statement text with mathematical notation preserved, and a list of dependency identifiers (other candidates that this statement requires to be understood). Dependencies are validated against the candidate ID set; references the LLM hallucinates are filtered. Crucially, we impose *no vocabulary constraint during extraction*: the LLM captures whatever statement content and type it finds most faithful to the paper's own language, with canonical-vocabulary consolidation deferred to Section 3.4.

D.3 Stage 3: Unfold

The dependency lists from the Structurer define a directed graph. To make each statement readable in isolation — the property that dense retrieval needs — we expand statements layer by layer using Kahn-style topological layering. The algorithm partitions the DAG into levels: layer $\ell = 0$ contains every statement with in-degree zero (no dependencies — typically the primitive assumptions and standalone definitions that anchor the paper); layer $\ell + 1$ contains every statement whose in-degree drops to zero after removing layers $0, \ldots, \ell$. Layers are processed in order. For each statement in layer ℓ, the LLM receives the target statement and the already-unfolded text of its *direct* (first-order) parents from earlier layers, and produces a self-contained restatement that inlines the relevant parent content so the target can be understood without consulting any other source. Layer-zero nodes are already self-contained and pass through without an LLM call. First-order expansion is the key design move: transitive context is inherited through the layer ordering rather than retrieved all at once, so the unfolded text stays concise and errors in one layer do not propagate unboundedly. Cycles — rare in practice but possible when the LLM misattributes a dependency — are broken by promoting the lowest-in-degree node and logging the break. The output is one self-contained text per statement, ready for embedding.

D.4 Per-paper visualisation

Each per-paper DAG can be inspected interactively before any corpus-wide step is run. A standalone visualiser (`matlas_replica/visualize.py`) loads the unfolded JSON for a paper and renders a Sugiyama-style layered diagram in a self-contained HTML page: layer 0 at the bottom, dependency edges drawn upward, type-coloured chips on each node (theorem in red, definition in blue, assumption in yellow, equation in cyan, algorithm in purple). Hovering a node reveals the raw statement text, its unfolded form, and its dependency list. The visualiser is the diagnostic of choice for spotting Locator misses (statements absent from the DAG) and Structurer mis-attributions (cycles or implausible parent edges) on a per-paper basis. The corpus-wide complement — a UMAP of the full embedding cloud across all papers — is not part of the per-paper extraction and lives in Section 5.2.

fig04_embedding_umap.pdf

Figure 5: UMAP projection of unfolded-statement embeddings across the OLG corpus, coloured by statement type. Each point is one extracted statement from one paper; proximity reflects embedding-space similarity. [DATA-DEPENDENT: regenerate from `matlas_replica/data/index/` after full 97-paper extraction.]

fig08_pipeline_trace.pdf

Figure 6: Five-agent pipeline trace on one held-out OLG specification. The graph feeds each stage; cross-stage checks are anchored on the same condition vocabulary.